

Foundation of Risk Management^{*}

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1 Course introduction

- Risk management in financial institution (with a focus on Banks)
- Banks are facing various risks, including:
 1. Market Risk (adverse variation in market factors)
 2. Credit Risk (e.g. a client defaults and cannot pay its debt)
 3. Counterparty risk (e.g. a counterparty cannot pay the payoff of a derivatives because of its default)
 4. Operational Risk (linked to human error, fraud, or any deficiency in the institution's operational system)
 5. liquidity risk (e.g. a company cannot pay a coupon because of a bad timing of its cash inflows and outflows)
 6. Legal risk (e.g. a counterparty cannot be forced to meet its obligations as a result of an inadequate contract)
 7. Climate risk (physical and transition risk e.g. impacting collateral asset or investment in stranded asset)

- Different actors: within the bank: traders (front-office), risk department (middle-office, model validation team), senior management, shareholders; regulator; clearing houses; ratings agency; data providers etc.

Part I

Market Risk

2 Market Risk Evaluation

- **Definition:** Market risk is the the risk of losses in on- and off-balance sheet risk positions arising from movements in market prices.

↪ Banks must identify risk drivers (factors) influencing market prices and their correlations.

↪ Banks must be able to (re)price their positions using independent data sources.

↪ Banks must self-control the risks they take.

2.1 Main Risk Drivers

2.1.1 Interest Rates

- Banks face several types of interest rate risk (IRR), both in the banking book (loans, deposits) and the trading book (derivatives, bonds).
- Change in interest rate may affect banks in different way. It changes
 - a [bank's earnings](#), by changing its net interest income and the level of other interest-sensitive income and operating expenses (ALM)
 - book value perspective: "How will the profit change over the next years?"
 - the underlying [economic value](#) of a bank's assets, liabilities, and off-balance sheet instruments (future cash flows must be "present-valued" or discounted at prevailing interest rates)
 - market value perspective: "How much value is lost immediately due to this rate move?"

Example of repricing mismatches (IR mismatches, gap risk): fixed rate loan guaranteed by short dated liabilities (deposits) whose IR cost is random.

Consider the simplified balance-sheet:

Assets	Liabilities
5-year fixed-rate loan	Overnight customer deposits
Notional: 100 million	Notional: 100 million
Fixed rate: 4%	Rate: 1%

↪ **Interest Rate Shock:** +1% => Assume a parallel upward shift of +1% in market rates.

- **Effect on NII - Book-Value perspective (Earnings View):**

- *Before the Shock:* The initial Net Interest Income (NII) is:

$$\text{NII}_0 = 4\text{m} - 1\text{m} = 3\text{m per year.}$$

- *After the Shock:*

- The loan remains fixed at 4%.

- Deposits reprice immediately from 1% to 2%.

Thus:

$$\text{NII}_1 = 4\text{m} - 2\text{m} = 2\text{m per year.}$$

The effect on earnings is:

$$\Delta\text{NII} = -1\text{m per year.}$$

Example of Market-Value Impact:

The 5-year fixed-rate loan is now seen as a bond with:

$$\text{Coupon} = 4\%, \quad \text{market rate at } t_0 = 4\%, \quad \text{market rate at } t_1 = 5\%.$$

- Present Value Before the Shock

At 4% discount rate, the bond is at par:

$$PV_0 = 100.$$

- Present Value After the Shock

The present value at 5% is:

$$PV_1 = 4 \cdot \frac{1 - (1.05)^{-5}}{0.05} + \frac{100}{1.05^5} = 95.67.$$

Thus the market-value loss is:

$$\Delta PV = 95.67 - 100 = -4.33.$$

- “Interest rate risk” is a term encompassing a number of different forms of risk.
 - Yield curve risk (“risk-free” rates): level/slope changes
 - Basis risk: Basis risk arises from imperfect correlation between changes in interest rates from different market sectors.

Example: a strategy of hedging a corporate bond portfolio with Treasury futures

- Optionality: standard IR derivatives (caps, floors, swaptions, etc), embedded options (bonds/notes with call or put provisions, loans that give borrowers the right to prepay balances).
- Risk free rates: "true" risk free rates are since the beginning of the 2020's the overnight rate: SOFR (USD), ESTR (euro), SONIA (£),... (LIBOR has been discontinued in 2020: it was no more considered risk free since 2008 and has been manipulated between 2005-2010...) - for longer maturities : some government bonds may be used

2.1.2 Foreign Exchange Rates

Foreign currency = any currency other than a bank's reporting currency.

FX (or currency) risk = the risk that the value of on- or off-balance sheet positions will be adversely affected by movements in exchange rates.

2.1.3 Credit Spread

- Credit Spread Risk is a specific IR risk that arises from a change in the market (credit) risk premia of some entities.

Crucial for bonds and credit derivatives valuation and risk management.

2.1.4 Equity Prices

Overall market movements that affect any stock (general or systematic risk) or from investing in a particular stock (specific or idiosyncratic risk).

Example: the bank sells call options on an Index

2.1.5 Commodity Prices

unexpected losses arising from a change in the market price of a commodity.

- Price risk in commodities is often more complex and volatile than price risk associated other factors : influence of natural events (floods and droughts), political uncertainties, supply and demand, seasonality...

2.2 Mark-to-Market (MtM) Valuation

- The process of calculating a market value for an asset is called “marking-to-market”.
- Purpose: to determine unrealized Profit and Loss (P&L) by comparing the current market value of a position against the value of the position at the time of the last mark-to-market or valuation date (usually the previous trading day), or entry to the transaction.
- Market risk exposures are mark-to-market values that are sensitive to the previous market drivers.
⇒ Market risk calculations rely on market value changes!
- The mid-market price (the average of the bid and offer quotes) is often used in performing the 'mark'.

2.2.1 Obtaining Data for Mark-to-Market Purposes

- A key step = the gathering of appropriate market prices or rates.
- Such data should be obtained from sources independent of traders or the front office.
↪ This ensures that a bank can revalue its positions on a conservative and cautious basis
- Independent market data providers: Thompson-Reuters, Bloomberg, Platts (part of S&P Global).
- Market prices/rates are also published on computer screens by various brokers and market makers.

- But, for many positions, obtaining prices or rates for mark-to-market purposes is not so straightforward.
 1. the market itself may not have a liquid or reliable quote available:
 - securities that trade infrequently (illiquid securities),
 - securities with complex structures or specialized risks that limit trading and liquidity (collateralized debt obligations or CDOs, structured derivative notes)
 - (non-standard) exotic options. They may be highly customized to meet client specifications.
 2. a position may be so large that closing it out may impact the market. In practice, when wanting to sell off a large position under normal market conditions, most banks sell it at a discount.

- To address the difficulty in obtaining data for MtM purposes:
 - [Marking-to-model](#): any valuation that is benchmarked, extrapolated, interpolated, or otherwise calculated from market inputs (calibration)
 - Base the MtM revaluation on the [quotes that are available](#) for the relevant instrument, but some issues potentially (reliability, moral hazard, conflict of interest...)
 - Use the [average of prices/rates](#) quoted by a set panel of dealing firms and brokers, after discarding outliers
 - [external providers](#), as Markit's Totem service (part of S&P Global now).

2.3 Sensitivity Analysis

- A mark-to-market valuation ensures the basis for translating observable market price changes into changes in the value of a bank's portfolio.
- The simplest form of market risk measurement : **sensitivity analysis**
- *Principle:* Apply a shift to a specific risk factor in order to assess the sensitivity of the portfolio to changes in that risk factor.

↔ Example: how much will the IR portfolio gain or lose if the government bond yield curve shifts up or down by one basis point from today's level?
- The calculation of a portfolio's sensitivity to changes in each of the risk factors in turn (sensitivity analysis) is simple.

- Consider a derivative, whose current market value is

$$V_t = u(S_t, t, \Sigma, R_t, \dots),$$

where

- S_t is a vector of m underlyings,
 - $\Sigma = [\sigma_i \sigma_j \rho_{ij}]_{1 \leq i, j \leq n}$ is a covariance matrix,
 - R_t is a q vector of (not risky) interest rates.
- Traders commonly use their so-called "Greeks" to evaluate the associated risk of this derivative:
 - the Delta (vector) $\partial u / \partial S_t$;
 - the Theta $\partial u / \partial t$;
 - the Vega (vector) $\partial u / \partial \sigma_i$, $i = 1, \dots, m$;

- the (matrix of) correlation sensitivities $[\partial u / \partial \rho_{ij}]$, $i \neq j$, $i, j = 1, \dots, m$;
- the Rho (vector) $\partial u / \partial R_t \dots$

- There are other Greeks, corresponding to higher order risks:
 - the Gamma (matrix) $\partial^2 u / \partial^2 S_t$;
 - the Vanna (matrix) $\partial^2 u / \partial S_{it} \partial \sigma_j$;
 - the Volga (matrix) $\partial^2 u / \partial \sigma_i \partial \sigma_j$, $i, j = 1, \dots$, etc.
- Greeks allow **hedging derivatives** (at least approximately).
- Necessary conditions to lead an efficient hedging strategy:
 - the availability of hedging instruments (!)
 - frequent adjustments of the hedging position
 - no jumps in the drivers of u (the "Greeks logic" = limited expansions)
 - reasonable transaction costs...

Example 2.1. We own a portfolio of 100 exotic options, whose value is $u(S_t, t, \sigma_t)$.

The delta of one of these options is $\delta_t = 0.7$, and its vega is $\nu_t = -2.1$.

We want to delta-vega hedge these options with the underlying S_t and with vanilla calls, whose own vega is $\nu_t^{(H)} = 1.4$ (their delta is considered as zero).

An (approximate) hedge of this portfolio is

1. to be short 70 underlyings;
2. to buy $100 * 2.1 / 1.4 \simeq 150$ vanilla calls.

At the first order, for the delta and vega hedge, we have that the variation of value is

$$\Delta u \simeq 100\delta_t\Delta S + 100\nu_t\Delta\sigma$$

Now, we do have that $\Delta u - 70\Delta S + 150\Delta C \approx 0$ since we assumed $\Delta C \simeq 1.4\Delta\sigma$ (neglecting the delta of the call or assuming the calls are already delta hedged).

A perfect hedge does not exist. In a Black-Scholes world, the underlying is assumed to follow a diffusion process

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma_t dW_t,$$

where (W_t) is a univariate Brownian motion (to simplify).

(σ_t) is the historical (or "realized") volatility process. Assume *no interest rates*.

Consider a vanilla european option with maturity T : Payoff at maturity is $g(S_T)$ with $g : \mathbb{R} \mapsto \mathbb{R}$. Its price depends on the level of volatility σ_I "implied" by the market.

According to the financial theory, this option can be delta hedged continuously in local volatility model given by σ_I .

The PNL generated by a delta-hedged strategy between t_1 and T is then

$$\text{Hedging Error}(t_1, T) = \int_{t_1}^T \frac{S_t^2}{2} \partial_{xx}^2 u(t, S_t) (\sigma_I^2 - \sigma_t^2) dt. \quad (2.1)$$

Hedging error comes also from discrete rebalancing.

Sketch of proof of (2.1): The product is priced and hedged using the following PDE:

$$\partial_t u + \frac{1}{2} \sigma_I^2 x^2 \partial_{xx}^2 u = 0$$

The hedging potfolio has dynamics for $t_1 \geq t$:

$$V_t = u_{t_1} + \int_{t_1}^t \delta_s dS_s$$

The hedging error at time T is $g(S_T) - V_T = u(T, S_T) - V_T$ that can be expanded using Ito's formula to get:

$$PNL_T = u_{t_1} - V_{t_1} + \int_{t_1}^T (\partial_x u(t, S_t) - \delta_t) dS_t + \int_{t_1}^T \partial_t u(t, S_t) + \frac{1}{2} \sigma_t^2 S_t \partial_{xx}^2 u dt$$

Note that $u_{t_1} = V_{t_1}$ and the option is indeed Delta-hedged so that

$$PNL_T = \int_{t_1}^T (\partial_t u(t, S_t) + \frac{1}{2} \sigma_t^2 S_t \partial_{xx}^2 u) dt = \int_{t_1}^T \frac{S_t^2}{2} \partial_{xx}^2 u(t, S_t) (\sigma_I^2 - \sigma_t^2) dt.$$

where for the last inequality we used the PDE satisfied by u . □

- Sensitivities are calculated intensively, intra-day, for every (traded) product.
- They are used
 - by traders, to manage their positions (at the level of the book generally);
 - by managers (head of desk, and above) to explain P&L variations;
 - by risk managers, to evaluate the riskyness of their positions and portfolios.

↔ Limits are placed on these various sensitivities

- However, [Sensitivity Analysis](#) suffers from a number of drawbacks:

- Local analysis only: it ignores large market move (jumps).

- It ignores interactions between factors:

- Examples:* the movements of different yield curves (same currency or not) are highly correlated over the medium-term because they are subject to the same broad economic and market factors.

- Spot and vol are also negatively correlated. etc.

- difficult to aggregate on large/complex portfolios (or at the bank level), false netting could appear.

- Example:* Is a USD 10 million loss due to a one basis point change in 10Y Bond yield a cause of more concern than a USD 5 million loss due to a one basis point change in the EUR/USD exchange rate?

- Sensitivities are model dependent...

⇒ need for additional and more sophisticated tools.

- use of portfolio models to capture correlation
- a statistical/historical approach to risk measurement (VaR,ES).
- calls for stress testing scenarios

2.4 Risk Monitoring and Control

- Objective of market risk management: to maintain a bank's market risk exposure within self-imposed parameters over a range of possible changes in market risk factors.
- One way of achieving this objective: to implement a detailed [system of market risk limits](#)
 - Limits can be set in various ways depending on the organization and complexity of the bank's activities.
 - Limits may be imposed on total VaR exposure, position size, or sensitivities of positions to risk factors (duration, convexity limits on bond positions, Greeks limits on options...).
 - Limits are imposed in a hierarchical fashion whereby portfolios are divided by trader, trading desk, business line or location, and product/portfolio.

- The definition of limits on market risk according to the bank's risk appetite

↔ This is the responsibility of the board of directors and senior management

- Meaningful thresholds are necessary to secure business for the bank and optimize the allocation of resources

too low limits compared (to a bank's activity) $\Rightarrow$ frequent limit breaches, profitable business transactions may be constrained.

too high limits $\Rightarrow$ suboptimal allocation of capital resources, fail to serve their ultimate purpose of safeguarding the bank from excessive market risk exposure.

- Limit exception (violation) should be made known to senior management without delay.

Banks should have a **clear policy** in place as to how senior management will be informed and what action should be taken in such cases.

Some limits are absolute (hard limits, that cannot be violated), when others can be breached for a short period of

time (soft limits).

- A system of market risk limits imposes upper boundaries on a bank's business volume, unless it is possible to hedge the risks

↔ **Hedging** = taking an action so that there will be an opposite position created to the risk position being hedged.

Hedging is usually accomplished by: buying/selling an instrument or entering into a contract that has the opposite market value exposure.

- The use of **Derivatives** = the main tool to hedge market risk.
 - to a bank changes a risk it does not want to manage into a risk it knows how to manage.
 - Derivatives alter risk exposures by resizing or reversing them, and enable market risk income to become less dependent on price and rate movements.

3 Value at Risk (VaR)

3.1 Risk measures

- Above, we have seen that sensitivities are a way of measuring risks (and controlling them) - see the drawbacks mentioned there also
- Standard deviation is also a natural risk measure:
 - it goes back to Markowitz [16] and is the cornerstone of classical portfolio theory => formalise the need for diversification
 - but it does not relate easily to the probability of loss (two distributions with same mean and variances may have totally different loss distribution)
 - it does take into account in the same way positive or negative deviation from the mean (only one side is of interest here however)

- Need for a risk measure that is simple (one number), can be easily interpreted and focus on "downside risk". VaR is a typical example.
- we will see that we can also ask for more from a risk measure (see coherent risk measure below... VaR is not a coherent risk measure)

3.2 Definition of Value-at-Risk (VaR)

- Introduced by the American bank JP Morgan at the start of the 1990s (development of Riskmetrics method and then Creditmetrics).
- The idea was to summarize the risk impacting a portfolio or an assets- and-liabilities position in a single measure with a direct interpretation.
- It can apply to an isolated position, a portfolio or to a whole bank.
- Value-at-Risk (VaR) is related to a random variable L , that is the loss incurred by a position on given period h . (**loss is positive**, gain is negative):

$$L = V_0 - V_h \tag{3.1}$$

with V_0 portfolio value today, V_h portfolio value at the end of the period.

- It is used to measure market risk (h=1 day, h= ten days) and credit risk (h=1 year).

3.2.1 Definition:

1. The usual VaR definition: at some level $\alpha \in [0, 1]$,

$$\mathbb{P}(L \leq VaR_\alpha) = \alpha, \quad (\text{for } L \text{ with positive p.d.f.})$$

and, more generally,

$$VaR_\alpha = \inf\{v \mid \mathbb{P}(L \leq v) \geq \alpha\}.$$

2. The shifted VaR (the unexpected loss in Credit Risk):

$$\overline{VaR}_\alpha = VaR_\alpha - E[L].$$

- note: $\alpha \geq 0.95$ in practice.
- notation: $F(t) = \mathbb{P}(L \leq t)$ c.d.f., F^{-1} : (generalised) inverse (left-continuous), $VaR_\alpha = F^{-1}(\alpha)$.

- What is $\mathbb{P}$ above? for risk management purpose $\mathbb{P}$ is the historical probability.

3.2.2 Properties

Remark 3.1. *i) A Value-at-risk calculation does not consider the order of magnitude of the (extreme) losses that will be suffered $1 - \alpha\%$ of times.*

ii) Let Z be a $\mathcal{N}(0, 1)$ gaussian random variable. Denote by $N(\cdot)$ its CDF.

Then, simply $VaR_\alpha[Z] = N^{-1}(\alpha)$.

*If $L = \mu + \sigma Z$, then $VaR_\alpha[L] = \sigma * N^{-1}(\alpha) + \mu$.*

$\hookrightarrow$ In a gaussian world, Var and $Std Dev$ are similar indicators.

- VaR is positively homogeneous: $Var[\lambda L] = \lambda Var[L]$, for $\lambda \geq 0$.
- VaR is translation invariant: $VaR[L + a] = Var[L] + a$, for $a \in \mathbb{R}$.
- Var is Monotone: $Var[L_1] \leq VaR[L_2]$ if $L_1 \leq L_2$.

But...

- VaR is not subadditive.

Example: Consider 100 iid bonds: 2%-coupon, 100 face value, 1% default probability (period= 1 year).

$$X_i = \begin{cases} -2 & \text{with probability 99\% (coupon is paid)} \\ 100 & \text{with probability 1\% (bond defaults)} \end{cases}$$

Two portfolios losses: $L_1 = \sum_{i=1}^{100} X_i$ (one of each bond) and $L_2 = 100X_1$ (100 bonds of type 1).

Note that $VaR_{95\%}(X_1) = \inf\{t | \mathbb{P}(X_1 \leq t) \geq 0.95\} = (-2)$.

$$VaR_{95\%}(L_1) = VaR_{95\%}\left(\sum_{i=1}^{100} X_i\right) > 0 > VaR_{95\%}(L_2) = \sum_{i=1}^{100} VaR_{X_i}$$

Hence the well-diversified portfolio L_1 gets a higher (VaR-)risk charge than the very concentrated portfolio L_2 !

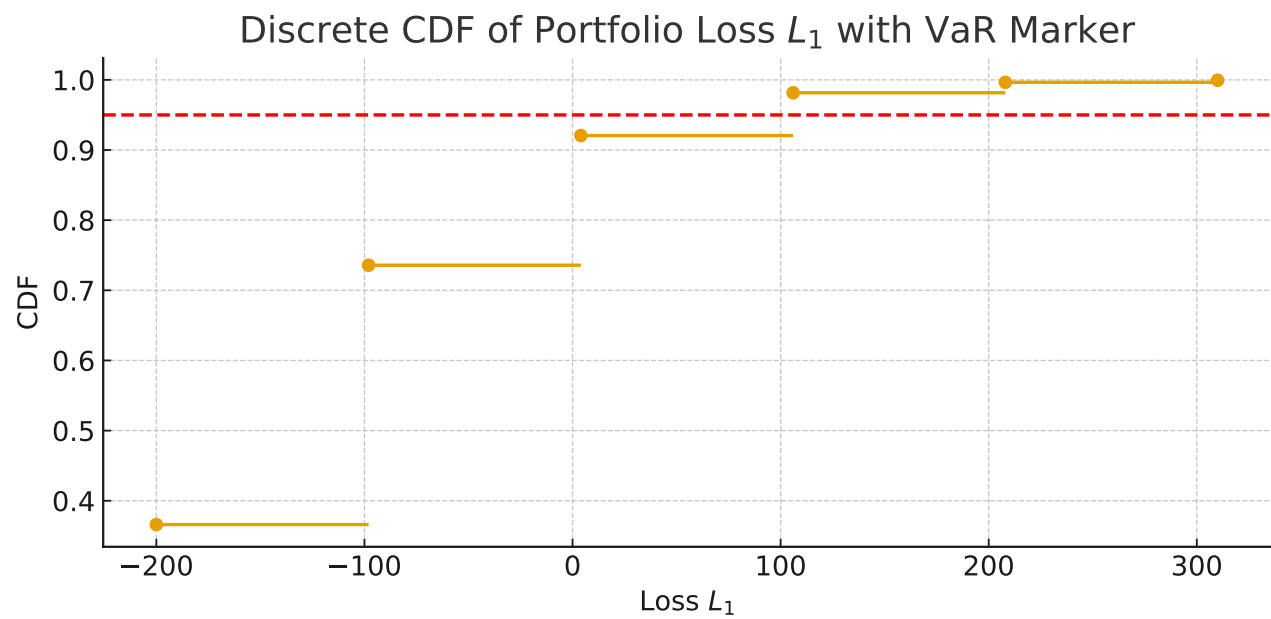


Figure 1: CDF of the portfolio loss L_1

3.2.3 Estimating VaR in practice

- In practice, we have to evaluate VaR through finite samples.

$\hookrightarrow$ we calculate estimated VaRs and not true VaRs.

- **Definition:** The *empirical VaR* is given by

$$\widehat{VaR}_\alpha := \inf\{x | F_n(x) \geq \alpha\},$$

where F_n is the empirical cdf of losses $(L_i)_{1 \leq i \leq n}$:

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(L_i \leq x).$$

$\hookrightarrow$ **In practice**, we order losses, and we find the boundary index k such that

$$\widehat{VaR}_\alpha = L_{(k)}.$$

Namely $k = \lceil n\alpha \rceil$.

Remark 3.2. *If the losses (L_i) of the portfolio are assumed independent and identically distributed and L_i has a density f . It can be proved [14] that*

$$\sqrt{n} \left(\widehat{VaR_\alpha} - VaR_\alpha \right) \xrightarrow{law} \mathcal{N} \left(0, \alpha(1 - \alpha) / f^2(VaR_\alpha) \right).$$

- comments on the definition:

(i) Let us consider the order statistics $L_{k:n} = L_{(k)}$, we thus have

$$F_n(x) = \sum_{k=1}^n \frac{k}{n} \mathbf{1}_{[L_{(k)}, L_{(k+1)}]} \quad (3.2)$$

with the convention $L_{(n+1)} = +\infty$. (associated to the random empirical measure $(\frac{1}{n} \sum_{k=1}^n \delta_{L_k})$)

$\hookrightarrow$ so for $\frac{k-1}{n} < \alpha \leq \frac{k}{n}$, $F_n(x) \geq \alpha \implies F_n(x) = \frac{k}{n}$ and $\widehat{VaR}_\alpha = k = \lceil n\alpha \rceil$

(ii) convergence holds true if there is an unique quantile at level α . see [1].

3.3 Computation of VaR

3.3.1 Historical VaR

- The simplest way. Several steps:
 - Recording of the [past values of the relevant market factors](#)
 - Calculation of the daily past returns of the chosen factors
 - For a one day (or ten days) horizon, calculation of the values of the portfolio, assuming these current positions are constant in time. $\Rightarrow$ a sample of values for this portfolio.
 - Calculation of the empirical quantile for this distribution of values. Note: Necessity of a [scaling factor \$\sqrt{10}\$](#) in the case of one-day calculations.

The implementation is simple and quick.

- It is possible to weight observations, so that the most recent observations have more influence on the loss distribution than the oldest ones.

Typically, for some constant $\lambda \in (0, 1)$, L_t being the estimated loss induced by the factor returns observed t days ago

$$\mathbb{P}(Loss_T \leq x) \simeq \frac{1 - \lambda}{1 - \lambda^m} \sum_{t=1}^m \lambda^{t-1} \mathbf{1}_{\{L_t \leq x\}},$$

- Beware of “[Ghost effects](#)”: VaR may be unduly high because of a past period of high volatility. This VaR will continue to be high until m days have passed (and then the historical observations have fallen out the sample period).
 $\Leftrightarrow$ The historical VaR depends on the past period of time that is considered ($\geq$ one year)

- A priori, the wider the period of time, the better the model.
- But very long data periods ($>$ several years) can induce their own problems:
 - aged data
 - persistent ghost effects

- the news in current market observations is likely to be drowned out by older observations

3.3.2 Parametric VaR

- Popular and relatively easy approach. Popularized by RiskMetrics.
- In a Gaussian world $\leftrightarrow$ Key building block is the construction of a **variance-covariance matrix** to calculate the value-at-risk.
- The variance-covariance approach to calculate VaR assumes:
 - the likely distribution of returns on financial assets in the future can be measured based on the observed volatility of the returns on those assets in the past
 - the distribution of those returns in the future will be **normal**
 - the correlations between assets are stable.

$\Rightarrow$ the knowledge of the means and the variance-covariance matrix of the market factors is sufficient to describe the randomness of the market.

Portfolios of basic assets

- The change in the value of a linear portfolio is a random variable

$$\Delta P_t = P_{t+1} - P_t = \sum_{j=1}^q m_{jt} r_j = \vec{m}_t' \vec{r}.$$

$$P_t = \sum_{j=1}^q m_{jt}.$$

$\vec{r} = (r_1, \dots, r_q)$ the (stationary) daily returns of every individual security

$\vec{m}_t = (m_{1t}, \dots, m_{qt})$ the t -market values of the q underlying positions.

- The usual basic assumption: the returns of the underlying securities in the portfolio are normal, or $\vec{r} \sim \mathcal{N}(\mu, \Sigma)$.

$$\Rightarrow \Delta P_t \sim \mathcal{N}(\vec{m}_t' \mu, \vec{m}_t' \Sigma \vec{m}_t) \equiv \mathcal{N}(\nu, \sigma^2).$$

- The “ α -quantile” of the loss (L) distribution at level α is defined by

$$\mathbb{P}(\text{Loss between } t \text{ and } t+1 \leq q_\alpha) = \mathbb{P}(-\Delta P_t \leq q_\alpha) = \alpha,$$

$$\Phi\left(\frac{q_\alpha + \nu}{\sigma}\right) = \mathbb{P}\left(\mathcal{N}(0, 1) \leq \frac{q_\alpha + \nu}{\sigma}\right) = \alpha.$$

- Actually, the Value-at-Risk is the amount of money that can be lost without having been anticipated: ν can be anticipated.

$\Rightarrow$ **Market VaR** is the quantile of the distribution of losses less the mean of this distribution ($= -\nu$).

$$UL_\alpha = q_\alpha + \nu = \sigma\Phi^{-1}(\alpha) \quad (UL \text{ stands for "Unexpected Loss".})$$

- In practice, $\nu \simeq 0$ on the small time interval (it is negligible compared to the standard deviation)
- If $\alpha = 0.99$ (resp. $\alpha = 0.95$) then $\Phi^{-1}(\alpha) = 2.33$ (resp. $\Phi^{-1}(\alpha) = 1.65$).
- Under the gaussian assumption, the parametric VaR is proportional to the volatility of the loss distribution

- Above, the asset returns are the market risk factors (one risk factor for one asset)
- It is also usual to consider linear factor models: asset returns are related to a set of risk factor $\vec{F}_t = (F_t^1, \dots, F_t^k)$ with

$$\vec{r} = B\vec{F} + \vec{\varepsilon}_t$$

with B the sensitivity matrix to the risk factors, $\vec{\varepsilon}$ the vector of idiosyncratic risks with $\mathbb{E}[\varepsilon_t] = 0$.

- Typical choice are the one-factor model (CAPM) of Sharpe [19] or the three-factor model of Fama & French [10].
- The Gaussian assumption of returns may be used for the most basic financial products or classical market factors.
- As already mentioned, the key point is to get the best estimate of the variance-covariance matrix for forecasting the loss distribution.
- The original approach implemented in RiskMetrics is Exponentially Weighted Moving Average (EWMA):

$$\widehat{\Sigma}_t = \lambda \widehat{\Sigma}_{t-1} + (1 - \lambda) \vec{r}_{t-1} \vec{r}_{t-1}^\top$$

Using a finite number of samples n , the estimator reads

$$\hat{\Sigma}_t = \sum_{s=1}^n \frac{1-\lambda}{1-\lambda^n} \lambda^{s-1} \vec{r}_{t-s} \vec{r}_{t-s}^\top$$

The original value in RiskMetrics was $\lambda = 0.94$: for $n = 250$ (days), this means that 50% of the weights corresponds to the 12 last days.

Portfolio of options

- When dealing with derivatives, the assumptions made previously are no longer valid: returns are not gaussian, not stationary:

- Example of a Call Option on asset S : C_0 is the value at time 0, the value at time h is a non-linear function ψ of S_h .

The Call Option returns is:

$$R_C = \frac{\psi(S_h(1 + r_h)) - C_0}{C_0}$$

$\hookrightarrow$ in practice, approximation of the option prices by some limited expansion w.r.t. their underlyings (market factors).

- Partial Valuation in a Linear Model: [Delta-Normal method](#).

- The portfolio value $P_t = p(t, Y_t^1, \dots, Y_t^M)$: p is a non-linear function of the risk factors $(Y^m)_{1 \leq m \leq M}$

- The Loss (L) between t and $t + h$ is approximated by

$$L = -(P_{t+h} - P_t) \simeq - \left(\partial_t p(t, Y_t) h + \sum_{m=1}^M \partial_m p(t, Y_t) (Y_{t+h}^m - Y_t^m) \right) \quad (3.3)$$

by a Taylor expansion.

- This allows to recover analytic formulae if the risk factor are Gaussian

Example: a call option on a no dividend-paying stock.

Its value at time 0 is

$$C_t = S_t \Phi(d_1(t, S_t)) - K \exp(-r(T-t)) \Phi(d_2(t, S_t)), \text{ with } d_1 = \frac{\ln(S_t/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Loss(L) between now (t=0) and h (next day):

$$L := -(C_h - C_0)$$

$\hookrightarrow$ The law of L is clearly a cumbersome non linear function of S_h .

- “Delta” approximation:

$$L \simeq -\delta_0(S_h - S_0) \implies \text{we set } L \simeq \hat{L} \text{ with } \hat{L} := -\delta_0 S_0 r_h.$$

where $r_h = \ln S_h - \ln S_0$ is the stock return on the period $[0, h]$, δ_0 the Delta at time 0.

- “Delta-Gamma” approximation:

$$\hat{L} \simeq -\delta_0 S_0 r_h - \frac{1}{2} \gamma_0 S_0^2 r_h^2.$$

(non-centered χ^2 distribution)

- Multidimensional case is also tractable for VaR calculations:

- For the whole portfolio: N_i options (or underlyings) of type i (possibly the underlying itself). The i -th option O_i written on the $f(i)$ -th underlying $U_{f(i)}$.

$$\begin{aligned}\Delta P_t &= \sum_{i=1}^m N_i \Delta O_{it} \simeq \sum_{i=1}^m N_i \delta_{it} \Delta U_{f(i),t} \\ &\simeq \sum_{i=1}^m N_i \delta_{it} U_{f(i),t} r_{f(i),t} \simeq \sum_{j=1}^q \tilde{m}_{jt} r_{j,t}\end{aligned}$$

$$\tilde{m}_{jt} = \sum_{i|f(i)=j} N_i \delta_{it} U_{f(i),t}.$$

$\hookrightarrow$ First order expansion: The law of ΔP_t is approximated by a Gaussian random variable.

- Similarly with second order expansions:

$$\Delta P_t \simeq \sum_{j=1}^q \tilde{m}_{jt} r_{j,t} + \sum_{j,k=1}^q \bar{m}_{jkt} r_{j,t} r_{k,t}.$$

The law of ΔP_t a non centered chi-squared distribution with q degrees of freedom.

- Explicit computations of quantiles and then VaR_α .
- In theory: It is possible to improve the approximation further (Taylor expansions), but explicit distributions are no longer available.
- The previous gaussian assumption is surely crude: Other parametric distributions are possible (fat tails, volatility clustering).
- Rather unusual among the current practices in the industry because they do not allow explicit and quick calculations of VaR:
practical choice = gaussian distribution

3.4 Monte-Carlo VaR

- To deal properly with the non linearities of valuations (contrary to the parametric VaR approach): the so-called “Monte-Carlo risk measurement”
- The principle:
 1. assume that the underlying market factors are following some joint statistical law
 2. [simulation \(\$M\$ times\) of the underlying factors](#) at the relevant time horizon
 3. [full repricing](#) of every instrument, for every simulated path
 4. calculation of the M simulated portfolio values
 5. calculation of some empirical quantiles of this distribution, and deduce VaR values.

- Monte-Carlo VaR provide accurate valuations for a potentially large set of scenarios.
- Nonetheless, it depends on the assumptions on the underlying distributions and on the quality of the pricers (step 3).
- Time calculations = strong limitation for such methodologies.
- Usually, $M = 10000$ random draws is considered as sufficient for a 99% VaR calculation
- When some complex instruments (american options, bermudean swaptions, etc) belong to the book, the repricing step can become very expensive.

3.4.1 Comparison of VaR Approaches

Parametric VaR – Strong Points	Parametric VaR – Weak Points
Computationally efficient	Assumption of normality of returns
Method valid asymptotically (Central Limit Theorem)	Loss of fat tails
Only Greeks required	Requires estimation of volatilities and correlations between returns
Easy to handle incremental VaR (see definition below)	Second-order expansion may be insufficient to capture option risk (e.g., exotic options)

Table 1: Pros and cons of Parametric VaR

Historical VaR – Strong Points	Historical VaR – Weak Points
No distributional assumption	Complete dependence on a particular historical data set
No need to estimate volatilities and correlations (nonparametric)	Cannot accommodate structural changes in market conditions
Captures fat tails and other extreme events	Slow to reflect increases in risk due to sudden market turbulence
Aggregation across markets is easy	Short data set may lead to biased and imprecise VaR estimates
Full portfolio revaluation	Not always computationally efficient for portfolios with complex options

Table 2: Pros and cons of Historical VaR

Monte-Carlo VaR – Strong Points	Monte-Carlo VaR – Weak Points
Can accommodate any distribution of risk factors	Outliers are not incorporated into the distribution, making it difficult to evaluate the likelihood of extreme losses (far in the tails)
Can model any complex portfolio	Computationally intensive

Table 3: Pros and cons of Monte-Carlo VaR

- Concerning the accuracy of Var estimates:
 - parametric : depends on the validity of the normality assumption and on the degree of portfolio optionality
 - historical: Good, provided the historical data is a reliable guide to future market behavior
 - Monte-carlo: generally the most accurate method, provided sufficient simulation runs are carried out.
- Simulation approaches tend to be the choice of the most sophisticated financial institutions due to:
 - the flexibility of simulation methods in dealing with the ever-increasing range of complex instruments,
 - the advent of more efficient computational techniques in recent years,
 - the falling costs in information technology.

- The variance-covariance approach might be the most appropriate method for many smaller firms, particularly when:
 - they do not have significant options positions
 - they prefer to outsource the data requirement component of their risk calculations to a company such as Risk-Metrics
 - significant savings can often be made by using outsourced volatility and correlation data

4 Beyond VaR: Expected Shortfall

- *VaR* has some limitations:
 - Technical/practical ones (heavy to compute, gaussian assumptions often used)
 - Conceptual ones (*VaR* is not subadditive)
 - *VaR* gives a probability for the occurrence of a loss in excess of the *VaR* calculated, but *VaR* does not take into account the severity of loss.

4.1 Expected Shortfall

Definition: Assume the loss L has a pdf and satisfies $\mathbb{E}[|L|] < +\infty$. Another important alternative risk measure: the [Expected shortfall](#) at level α , defined as

$$\text{ES}_\alpha(L) = E[L \mid L \geq \text{VaR}_\alpha].$$

some remarks:

- The Expected Shortfall ES thus expresses the mean size of the losses above the VaR
- Sometimes it is called *Conditional VaR* or *tail VaR* or *expected tail loss* or *average value at risk*
- It has better theoretical properties than VaR (see coherent risk measures below).
- In 2012, the Basel Committee has promoted Expected Shortfall at level 97.5% to replace Value-At-Risk at level 99%, to calculate capital requirements... but the implementation has been delayed (part of the FRTB) to 2027?, 2030?

4.1.1 More on expected shortfall

- Variational representation: Let $X \in L^1$ be an integrable loss random variable and let $\alpha \in (0, 1)$. The Expected Shortfall at level α admits the following variational representation:

$$\text{ES}_\alpha(L) = \inf_{t \in \mathbb{R}} \left\{ t + \frac{1}{1 - \alpha} \mathbb{E}[(L - t)_+] \right\}, \quad (4.1)$$

where $(x)_+ := \max\{x, 0\}$ denotes the positive part.

- Integral representation

$$\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(L) \, du. \quad (4.2)$$

- When the distribution of L has a singular part (e.g. discrete random variable) : Acerbi and Tasche (2002) have defined the generalized expected shortfall by

$$\text{GES}_\alpha(L) := \frac{1}{1 - \alpha} \mathbb{E}[L \mathbf{1}_{\{L \geq \text{VaR}_\alpha(L)\}}] - \frac{\text{VaR}_\alpha(L)}{1 - \alpha} \left(\alpha - \mathbb{P}(L < \text{VaR}_\alpha(L)) \right).$$

This measure is (always) coherent (see definition below).

In practice: Let (L_i) be the loss. An empirical Expected Shortfall at level α may be

$$\widehat{ES}_\alpha := \frac{1}{n(1-\alpha)} \sum_{i=1}^n L_i \mathbf{1}_{\{L_i \geq \widehat{VaR}_\alpha\}}.$$

Alternatives:

$$\widehat{ES}_\alpha := \frac{1}{n - \lceil n\alpha \rceil} \sum_{i=\lceil n\alpha \rceil}^n L_{(i)},$$

$$\widehat{ES}_\alpha := \frac{1}{1-\alpha} \int_\alpha^1 F_n^{(-1)}(u) du,$$

where F_n is the empirical cdf and F_n^{-1} its generalised inverse (empirical quantile)

$$\widehat{ES}_\alpha := \frac{1}{n(\beta - \alpha)} \sum_{i=\lceil n\alpha \rceil}^{\lfloor n\beta \rfloor} L_{(i)}, \quad \beta \gg \alpha, \quad \text{see [20], etc.}$$

- *Monte-Carlo methods*: same as above but also stochastic approximation methods see e.g. [4, 8]

- *Parametric approach*: When L is assumed to be a gaussian $\mathcal{N}(\mu, \sigma^2)$ with p.d.f. denoted φ and c.d.f. denoted Φ , one computes

$$\text{ES}_\alpha[L] = \mathbb{E}[L | L \geq \text{VaR}_\alpha[L]] = \frac{1}{1 - \alpha} \int_{F^{-1}(\alpha)}^{+\infty} x \varphi(x) dx. \quad (4.3)$$

$$= \dots = \mu + \frac{\varphi(\Phi^{-1}(\alpha))}{1 - \alpha} \sigma. \quad (4.4)$$

- In this case, ES_α is also a standard-deviation based risk measure.
- When $\alpha = 0.975$, $\frac{\phi(\Phi^{-1}(\alpha))}{1-\alpha} \approx 2.34\dots$

4.2 Coherent risk measures

see e.g. [2, 3].

Definition 4.1 (Coherent Risk Measure). *Let $\rho : \mathcal{X} \rightarrow \mathbb{R}$ be a mapping that assigns a real number to each (random) loss $X \in \mathcal{X}$. The risk measure ρ is called coherent if it satisfies the following *four axioms* for all losses X, Y and all $\lambda \geq 0, a \in \mathbb{R}$:*

1. Monotonicity

$$X \leq Y \quad a.s. \quad \Rightarrow \quad \rho(X) \leq \rho(Y).$$

A position with smaller losses should not have larger risk.

2. Subadditivity

$$\rho(X + Y) \leq \rho(X) + \rho(Y).$$

Diversification should not increase risk.

3. *Positive Homogeneity*

$$\rho(\lambda X) = \lambda \rho(X).$$

Risk scales proportionally with the size of the position.

4. *Translation Invariance*

$$\rho(X + a) = \rho(X) + a.$$

Adding a certain loss a increases risk by exactly a .

Example 4.1. *Let $u \mapsto \omega(u)$ be a non negative non-decreasing function then*

$$\rho(L) := \int_0^1 \omega(u) \text{VaR}_u(L) \mathrm{d}u$$

is a coherent risk measure.

Exponential spectral risk measure: $\omega(u) = e^{-\gamma(1-u)}$ for $\gamma > 0$.

Expected Shortfall: $\omega(u) = \frac{1}{1-\alpha} \mathbf{1}_{\{u \geq \alpha\}}$

ES is a coherent risk measure: we need to prove subadditivity.

Proof. We consider two random variables X and Y and denote $Z = X + Y$. To simplify, we assume that X , Y and Z have a p.d.f. We denote $\text{VaR}_\alpha[X] = x_\alpha, \text{VaR}_\alpha[Y] = y_\alpha$ and $\text{VaR}_\alpha[Z] = z_\alpha$. We compute then

$$(1 - \alpha) (\text{ES}_\alpha[X] + \text{ES}_\alpha[Y] - \text{ES}_\alpha[Z]) = \mathbb{E}[X(\mathbf{1}_{\{X \geq x_\alpha\}} - \mathbf{1}_{\{Z \geq z_\alpha\}})] + \mathbb{E}[Y(\mathbf{1}_{\{Y \geq y_\alpha\}} - \mathbf{1}_{\{Z \geq z_\alpha\}})]$$

Denote $\varepsilon = \mathbf{1}_{\{X \geq x_\alpha\}} - \mathbf{1}_{\{Z \geq z_\alpha\}}$, we have that

$$\mathbb{E}[X(\mathbf{1}_{\{X \geq x_\alpha\}} - \mathbf{1}_{\{Z \geq z_\alpha\}})] = \mathbb{E}[X\varepsilon\mathbf{1}_{\{X \geq x_\alpha\}}] + \mathbb{E}[X\varepsilon\mathbf{1}_{\{X < x_\alpha\}}]$$

we remark that $\varepsilon \geq 0$ on $\{X \geq x_\alpha\}$ and $\varepsilon \leq 0$ on $\{X < x_\alpha\}$. Thus,

$$\mathbb{E}[X(\mathbf{1}_{\{X \geq x_\alpha\}} - \mathbf{1}_{\{Z \geq z_\alpha\}})] \geq \mathbb{E}[x_\alpha\varepsilon\mathbf{1}_{\{X \geq x_\alpha\}}] + \mathbb{E}[x_\alpha\varepsilon\mathbf{1}_{\{X < x_\alpha\}}] = x_\alpha\mathbb{E}[\varepsilon]$$

But $\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbf{1}_{\{X \geq x_\alpha\}}] - \mathbb{E}[\mathbf{1}_{\{Z \geq z_\alpha\}}] = (1 - \alpha) - (1 - \alpha) = 0$ and

$$\mathbb{E}[X(\mathbf{1}_{\{X \geq x_\alpha\}} - \mathbf{1}_{\{Z \geq z_\alpha\}})] \geq 0$$

Same holds true for the second part with Y . We do get $\text{ES}_\alpha[X] + \text{ES}_\alpha[Y] \geq \text{ES}_\alpha[Z]$. □

4.3 Marginal risk measure - risk contribution

- What is the contribution of each asset/subset of assets to the overall VaR or ES of a given portfolio?

They are useful as measures for:

- identifying the major sources of overall risk
- analyzing the contribution of a collection of assets, such as stocks in a particular sector, to overall VaR
- assessing the impact on overall VaR of adding/removing an asset from a portfolio
- making hedging decisions
- providing information to monitor position limits
- analyzing the performance of individual traders or departments within a bank.

- A naive approach: a [stand-alone capital](#), i.e. a single VaR or ES per sub-portfolio (business line, desk or trader)...
but this may be completely off.

4.3.1 Portfolio approach

- Assume that the loss $\mathfrak{L}$ can be decomposed as

$$\mathfrak{L} := \sum_{i=1}^n L_i$$

The L_i 's are the underlying losses

- Say the various components of a portfolio, namely:

- $V_0 = \sum_{i=1}^n x^i$ ($x_i \in \mathbb{R}$ amount invested in asset i), $V_h = \sum_{i=1}^n x^i(1 + R_i)$ (R_i return of asset i).

$x^i = X_0^i = q^i \times S_0^i$ (q^i quantity of asset i hold constant on the period, S^i price of asset i), $X_h^i = q^i \times S_h^i$ and

$S_h^i = S_0^i(1 + R^i)$

- Then, $\mathfrak{L} = V_0 - V_h = \sum_{i=1}^n -x^i R_i$ and $L^i = -X_0^i R_i$

- consider a Risk measure $\mathcal{R}(\cdot)$ with the property of [positive homogeneity](#) (see above).

$\hookrightarrow$ For instance: Volatility of returns, Expected Loss: $E[\mathfrak{L}]$, $VaR_\alpha[\mathfrak{L}]$, Expected Shortfall: $E[\mathfrak{L}|\mathfrak{L} > VaR_\alpha]$...

- Then, $\mathcal{R}(\mathfrak{L}) = \mathcal{R}(L_1, \dots, L_n) = \tilde{\mathcal{R}}(x^1, \dots, x^n)$. Question: what is the contribution of L_i to $\mathcal{R}(\mathfrak{L})$?

Marginal contribution: Using Euler's theorem, one has

$$\mathcal{R}(\mathfrak{L}) = \sum_{i=1}^n \frac{\partial \tilde{\mathcal{R}}}{\partial x_i}(x^1, \dots, x^n) x^i$$

$\hookrightarrow C^i(x) := \frac{\partial \tilde{\mathcal{R}}}{\partial x_i}(x^1, \dots, x^n) x^i$ is the contribution of asset i to the total risk.

- This is very satisfying: **total risk measure = the sum of its marginal risk measures.**

Proof. Assume that $\tilde{\mathcal{R}} : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable. Then,

$$\tilde{\mathcal{R}}((1 + \epsilon)x) = \tilde{\mathcal{R}}(x) + \epsilon \sum_{i=1}^n \frac{\partial \tilde{\mathcal{R}}}{\partial x_i}(x^1, \dots, x^n) x^i + \epsilon \phi(\epsilon)$$

with $\phi(\epsilon) \rightarrow 0$ as ϵ goes to 0.

But by homogeneity, $\tilde{\mathcal{R}}((1 + \epsilon)x) = (1 + \epsilon)\tilde{\mathcal{R}}(x)$ which leads to:

$$\epsilon \tilde{\mathcal{R}}(x) = \epsilon \sum_{i=1}^n \frac{\partial \tilde{\mathcal{R}}}{\partial x_i}(x^1, \dots, x^n) x^i + o(\epsilon)$$

the proof is concluded by dividing by ϵ in the above equality and letting $\epsilon \rightarrow 0$. □

- Computation of $\frac{\partial \tilde{\mathcal{R}}}{\partial x_i}$ is generally difficult.
- Computations can be made in the case of parametric VaR.
- Example: The case of the gaussian framework:

Indeed, let x_j be the notional amounts invested in each market factors:

$$VaR_{para} = \sigma \Phi^{-1}(\alpha),$$

$$\sigma^2 = \sum_i x_i^2 \sigma_i^2 + \sum_{i,j,i \neq j} \sigma_i \sigma_j \rho_{ij} x_i x_j,$$

where $\Sigma = [\sigma_i \sigma_j \rho_{ij}]$ is the variance-covariance matrix of the underlying market factors

We deduce the marginal VaR of k :

$$\begin{aligned}
C^k(x) = \textit{Marginal VaR}_k &= \frac{\Phi^{-1}(\alpha)}{2\sigma} x_k \{2x_k \sigma_k^2 + 2 \sum_{j,j \neq k} \sigma_k \sigma_j \rho_{kj} x_j\}, \\
&= \frac{\Phi^{-1}(\alpha)}{\sigma} \{x_k \sigma_k^2 + \sum_{j,j \neq k} \sigma_k \sigma_j \rho_{kj} x_j\}
\end{aligned}$$

Incremental risk:(discrete marginal contribution) It is defined as the difference between the risk of the portfolio estimated with the exposure x_i and without it. namely:

$$\hat{C}^i(x) = \mathcal{R}(x_1, \dots, x_i, \dots, x_n) - \mathcal{R}(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_n)$$

- Importantly,

$$\sum_{i=1}^n \hat{C}^i(x) \neq \mathcal{R}(x).$$

Summary:

- To evaluate the risk of a particular asset/desk/business line in the context of a diversified portfolio : marginal or incremental measures

5 Regulatory viewpoint - Market Risk

- The Basel Committee for Banking Supervision¹ has been created in 1974.
- As of 2019 has 45 members from 28 jurisdictions, consisting of central banks and authorities with responsibility of banking regulation
- It agrees on standards for bank capital, liquidity and funding. Its recommendations are non binding (but generally made legally binding by local jurisdictions)
- This "master" regulator is located within the BIS (Bank for International Settlement)
- The Committee has issued the various Basel Accords.

¹https://en.wikipedia.org/wiki/Basel_Committee_on_Banking_Supervision

5.1 Original Basel I

- The 1988 Basel Accord² focused primarily on credit risk introducing the Cooke Ratio:

$$\text{Cooke Ratio} = \frac{\text{Regulatory Capital}}{\text{Risk-Weighted Assets (RWA)}} \geq 8\%.$$

- Risk-weighted assets is a measure of the exposure at risk: it is obtained by applying fixed regulatory weights to the exposure for broad asset classes e.g. 0% for OECD sovereign exposures, 50% for residential mortgages, and 100% for corporate loans.

$$\text{RWA} = \sum_i \text{Risk weight}_i \times \text{Exposure}_i$$

²<https://www.bis.org/publ/bcbasc111.htm>

- Rationale [12]:

1. The Cooke ratio emerged in response to several weaknesses revealed during the banking crises of the 1970s and 1980s (notably the Latin American debt crisis and the U.S. Savings and Loan crisis).
2. The Basel I Accord aimed to:
 - (a) strengthen the solvency of internationally active banks,
 - (b) harmonise capital standards across jurisdictions,
 - (c) reduce competitive distortions arising from national regulatory differences.
3. the 8% threshold was not derived from a formal risk model: large international banks typically exhibited capital ratios between approximately 7% and 9% when measured on a risk-weighted basis.
↔ it was a political compromise.

5.2 Basel I/II

- In 1996 An amendment to the Basel I³ was produced to incorporate market risk. (the Barings Bank collapsed in 1995)
- It also promoted two ways of measuring risk:
 - using standardized approach
 - using Internal Model
- The BASEL II Accords⁴ were published in 2004 implemented in 2008. Introducing
 - Operational Risk
 - A three Pillars concept. Pillar 1: minimum capital requirement, Pillar 2: supervisory review (capital add-ons, stress testing), Pillar 3: market discipline (disclosure requirements).

³<https://www.bis.org/publ/bcbs24.htm>

⁴<https://www.bis.org/publ/bcbs107.htm>

5.2.1 Market Risk: example of standardized approach

- The capital charge $\mathcal{K}$ for market risk is again $\mathcal{K} = E \cdot K$, with E exposure and K capital weight.
- E.g. for equity risk:
 1. specific risk: $K^s = 8\%$ for each position (idiosyncratic)
 2. general risk: $K^m = 8\%$ for the net position (market)
- Example: \$100M short *S&P500* and \$60M long *Nvidia*:

$$\mathcal{K}^s = 8\%|-100| + 8\%60 = 12.8 \text{ and } \mathcal{K}^m = 8\%|-40| = 3.2 \quad (5.1)$$

total charge = \$16M.

- There are special charges for IR risk, FX risk, commodity risk and option's market risk too.

- Option's market risk can be dealt with the *delta-plus method* using sensitivities (Delta, Gamma & Vega).
- The sensitivities can be computed analytically (e.g. Black-Scholes) or via bumping.
- Capital charge:

- The directional exposure is computed through the Delta.

For all options on asset i exposures are computed via

$$E_i = \left(\sum_j N_j \Delta_j \right) S_i \quad (N_j : \text{number of options of type } j)$$

The capital weight is then (as above) considered for specific and general risk to obtain

$$\mathcal{K}^\delta = K^s \sum_i |E_i| + K^m \left| \sum_i E_i \right| \quad (K^m = K^s = 8\%)$$

- We also have, for options on asset i ,

$$\mathcal{K}_i^\gamma = \frac{1}{2} \left(\sum_j N_j \Gamma_j \right) (S_i K_i^\gamma)^2 \quad \text{and} \quad \mathcal{K}_i^v = \sum_j N_j \mathcal{V}_j (\Sigma_j 25\%)$$

for equity $K_i^\gamma = 8\%$.

The shift (in the first order and second order expansion of the option price) is thus $\Delta S_i = 8\% S_i$ for Delta, $\Delta S_i = K_i^\gamma S_i$ for Gamma and $\Delta\sigma = \Sigma_j 25\%$ for vega.

5.2.2 Regulatory Parameters for internal method:

- The Basel Committee has specified some [minimum standards](#) for the purpose of calculating Market Risk capital requirements by the internal method, including:
 - Current portfolios/positions are “frozen” during the holding period (the risk horizon).
 - VaR must be computed on a daily basis.
 - A one-tailed confidence interval of 99% must be used.
 - The minimum holding period should be 10 trading days
 - The minimum historical observation period should be one year.
 - To update VaR models, banks should refresh their data sets at least once every three months.
 - Banks are permitted to recognize correlations within broad risk categories and possibly across broad risk factor categories.

- Banks' internal models are required to accurately capture the optional risk.
- The model used for calculating regulatory capital has to be the same one used for day-to-day internal risk management

- Capital charge of date t :

- Compute $\text{VaR}_t = \sqrt{10} \text{VaR}_{0.99}[\text{one day loss between } t \text{ and } t+1]$

- Required capital on day t :

$$\mathcal{K}_t = \max \left(\text{VaR}_{t-1}, \frac{m}{60} \sum_{i=1}^{60} \text{VaR}_{t-i} \right)$$

$m = 3 + \xi$ (with $\xi \in [0, 1]$) is a factor trying to capture model risk (ξ is determined by the regulatory backtests).

In normal period, $\text{VaR}_{t-1} \simeq \text{VaR}_{t-i}$ and then

$$\mathcal{K}_t = \mathcal{K}_{t-1} + \frac{m}{60} (\text{VaR}_{t-1} - \text{VaR}_{t-61})$$

and the impact of the new VaR_{t-1} is quite limited. But if there is a shock ($\text{VaR}_{t-1} \gg \mathcal{K}_{t-1}$), it is persistent for 60 days.

- The qualitative criteria set by the Basel Committee include:
 - A bank should have an independent risk control unit
 - The duties of this risk control unit include the generation and analysis of daily risk reports from the risk management system, and an assessment of the relationship between risk exposure and trading limits.
 - The risk control unit should undertake regular backtesting of the bank's risk models.
 - The board of directors and senior management should be actively involved in the risk control process.
 - The bank's internal risk measurement model must be closely integrated into the day-to-day risk management process of the bank.

Stress testing:

- Stress testing should be used as a supplement to the day-to-day risk analysis based on the bank's risk measurement model.
- Compliance with a documented set of internal policies, controls and procedures concerning the risk measurement system.
- A regular independent review of the risk measurement system by the bank's own internal auditing process. Furthermore, a review of the overall risk management process should take place at least once a year.

Backtesting:

- Basic idea: verifying that the internal model is consistent with a 99% confidence level.
- Based on a one-day holding period, it compares the computed VaR: VaR_{t-1} with the realized P&L: $L_t := -(P_t - P_{t-1})$ ⁵
- An exception occurs when $L_t > \text{VaR}_{t-1}$. Depending on the number of exceptions on a given period, the regulator determines ξ (the penalty term in the VaR formula above). Example of a 250 trading day period
 - "Green zone", 0 to 4 exceptions, $\xi = 0$
 - "Red zone", 10 plus exceptions, $\xi = 1$
 - "Yellow zone", 5 to 9 exceptions, ξ ranges from 0.4 to 0.85.

⁵Have in mind that losses are positive. P_t, P_{t-1} are Mark-to-Market values.

5.3 Basel III

- Developed as an answer to the deficiencies in financial regulation revealed by the 2008 financial crisis a.k.a Great Financial Crisis (GFC).
- The Basel III requirements were published by the Basel Committee on Banking Supervision in 2010⁶.
- For market risk, the Fundamental Review of the Trading Book (FRTB) was published and revised between 2013 and 2019. (not completely implemented in all countries)
- It takes into account Counterparty risk, simplify the computations of regulatory capital for operational risk, and add many other things. (See the first part of the course for an overview of Basel III and its implementation in Europe.)
- The final Basel III⁷ reforms was published in 2017. It has been extended several times and will be implemented gradually by 2028.

⁶<https://www.bis.org/publ/bcbs189.htm>

⁷<https://www.bis.org/bcbs/publ/d424.htm>

5.3.1 FRTB in a nutshell

↪ FRTB⁸ has still two approaches.

1. a standardized method (SA-TB). It is the sum of three main components:

- "general" market risk: a sensitivity-based method but considering more risk classes, more risk factors and a slightly different methodology than for Basel I/II. (a lot more detailed)
 - Default Risk Capital (DRC): it captures the jump to default risk (replacing the specific risk of Basel I/II)
 - Residuals risk add-ons: for risks difficult to measure in practice e.g. instrument with an exotic underlying (weather, natural disaster, longevity etc.)
- Note tha the SA-TB method is important it sets a 72.5% capital floor requirement that the IMA method must satisfy.

⁸<https://www.bis.org/bcbs/publ/d457.pdf> (only 136 pages!)

2. An internal model-based approach (IMA)

- A novelty compared to Basel I/II is the introduction of "trading desk" (a group of traders implementing a well defined business strategy and operating within a clear risk management structure).

(e.g. FX options trading desk, Interest Rate Swap trading desk).

may not reflect the original organisation of the bank... takes time to implement.

- The IMA are implemented at the level of trading desk. Some desk will be allowed the IMA, others won't (and use the standardized method)
- For the internal model-based approach the VaR at 99% is replaced by the ES at 97.5% confidence level.
- It takes into account liquidity class (the 10 day period holding is not valid for all instruments)

Part II

Credit Risk

- The overarching goal of this part is to understand the Basel III IRB capital requirement formula:

$$K = \sum_i \text{EAD}_i \text{LGD}_i \left[\Phi \left(\frac{\Phi^{-1}(\text{PD}_i) + \sqrt{\rho_i} \Phi^{-1}(0.999)}{\sqrt{1 - \rho_i}} \right) - \text{PD}_i \right] \quad (5.2)$$

- It is also known as the [Credit VaR](#).
- To achieve this we are going to:
 - define credit risk
 - describe some products bearing credit risk
 - present (basic) model to price (single name) product in credit risk framework
 - present a portfolio model to understand the above formula

6 Introduction

6.1 Components of credit risk

- Let us consider the following definition of *Credit Risk*[5] .

Definition 6.1. *Credit risk embedded in a financial transaction is the risk that at least one of the parties involved in the transaction will suffer a financial loss due to default or decline in the creditworthiness of the counterparty to the transaction or, perhaps, of some third party.*

- There are two components to credit risk:
 1. *Change in credit quality* (credit rating): For a given financial transaction/product, the risk of changes in value due to unexpected changes in credit quality of a counterparty or issuer.

Example 6.1. *A holder of a corporate bond bears a risk that the market value of the bond will decline due to decline in credit rating of the issuer. Though the issuer is still honouring his payment obligation.*

2. *Default risk*: Failure of honouring a payment obligation by the obligor.

Example 6.2. *Firms might default on principal or interest payments on loans or bonds, on tax obligations, on wages of their employees, or on invoices of their business partners, etc.*

Remark 6.1. (i) *Default Risk* is a drastic change in credit quality of the obligor...

(ii) Bankruptcy codes specify the obligor must honour all his payment obligations. A default on one obligation entails a default on all other obligations, so obligor cannot choose which claims he honours.

This makes possible to speak of the default of the obligor itself.

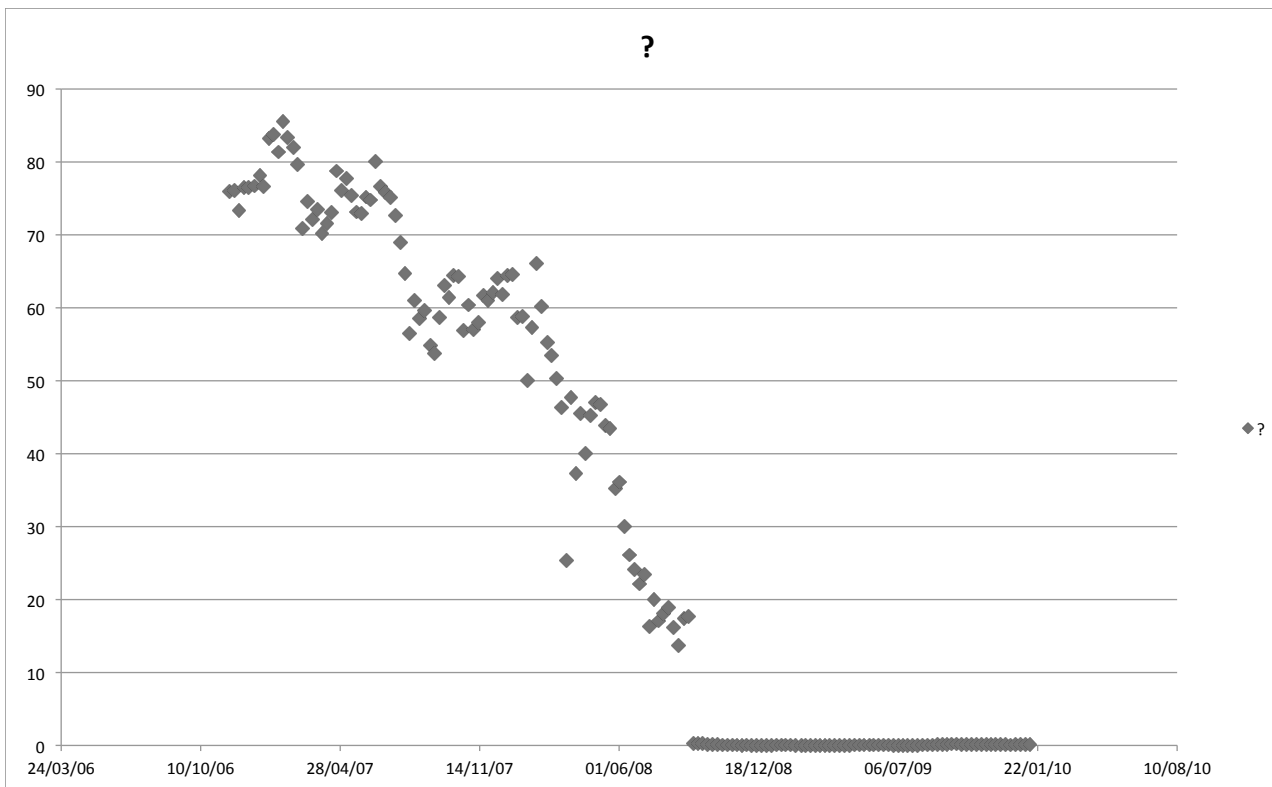


Figure 2: What does this chart represent?

There are other risks associated with default risk:

1. *arrival risk*: uncertainty whether a default will occur or not within a time horizon (usually one year).

probability of default (PD): $\mathbb{P}(\text{time of default} \leq \text{one year})$.

2. *timing risk*: uncertainty about the precise time of default

$\hookrightarrow$ measured by the *probability distribution function of the time of default*.

$\mathbb{P}(\text{time of default} \leq T), \forall T$.

3. *Recovery risk*: It reflects the severity of the losses if a default has happened.

Example 6.3. *For Loans and Bonds, it is expressed as a recovery rate, which is a percentage of the notional amount of the transaction.*

In practice it depends on the seniority of the debt and is quite difficult to assess.

In basic model, it is deterministic and set to 40%.

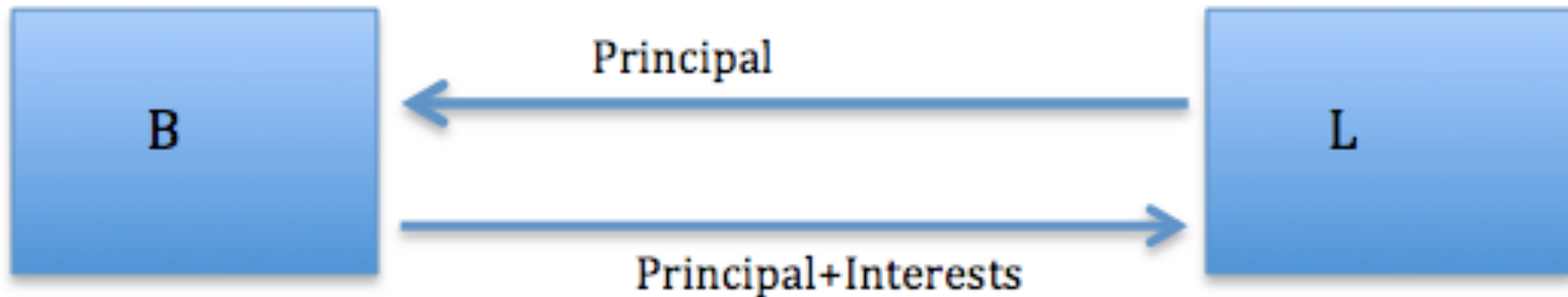
6.2 Example of assets bearing credit risk

- Any financial transaction bears a credit risk according to Definition 6.1!
- **Primary assets:** Loans and Bonds

Loans: oldest type of payment obligation & important ways to raise investment capital.

Loans are bilateral contract between the borrower (B) and the lender (L) (usually a bank). In its classic form, a loan consists of:

- a *principal* (or *notional amount*) which is lent to the borrower at origination i.e. the beginning of the transaction. This sum of money has to be repaid at the *maturity date*, i.e. the end of the transaction.
- *Interests*, which have to be paid regularly by the borrower to the lender.



↔ There are many variations to this basic structure especially because loans are bilaterally negotiated contract.

↔ But 'special features' will make loan difficult to transfer to other lenders.

Example 6.4. (i) *Loans to buy an house, to study etc.*

(ii) *Loans to raise capital to invest in a project etc.*

Bonds: securitised version of loans.

⇨ They are tradable: investors can buy bond after their origination and sell them before maturity.

⇨ The most important properties of a bond are its issuer, the coupon size and frequency (interests), the date of maturity and the notional.

⇨ Defaultable bonds prices contain a lot of information about the market's assessment of the issuer's credit risk (see below).

⇨ Bonds prices are also affected by interest rates change: so they bear an 'interest rate change' risk besides a credit risk.

For the lender :

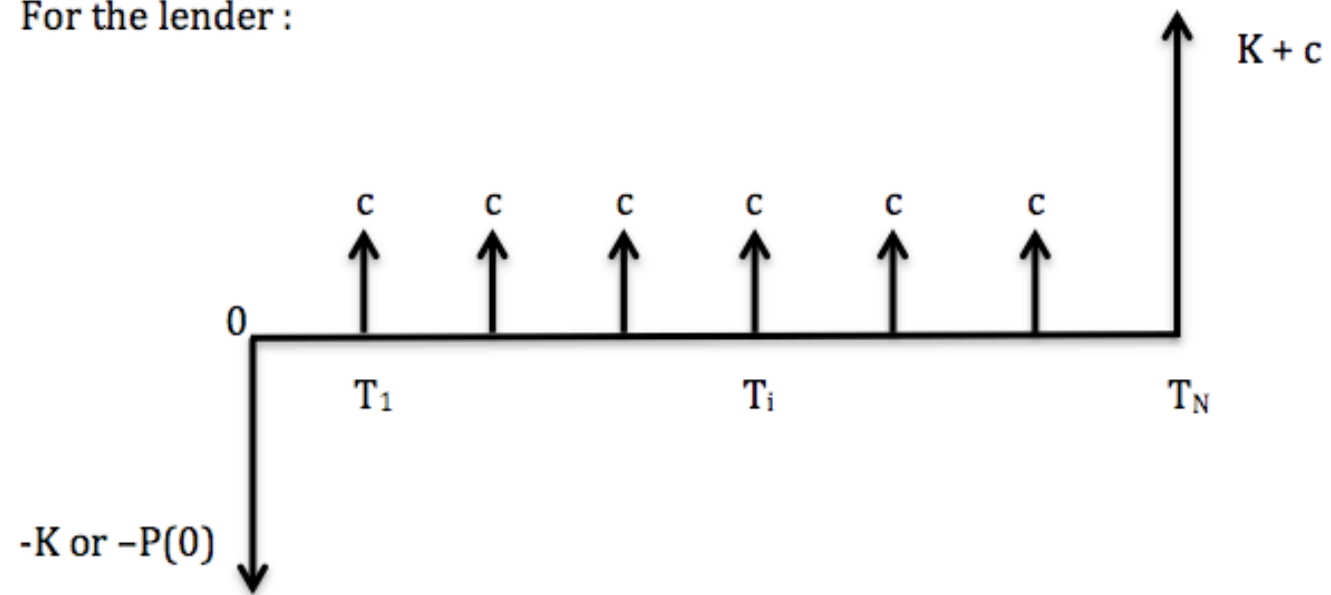


Figure 3: Example of a fixed-coupon bond

Zero-Coupon Bond (ZCB)= The most simple bond, does not pay coupon between origination and maturity.

At maturity, the nominal plus the interests are paid to the owner of the bond (the lender).

↔ Building blocks for credit risk modelling and the pricing of more complex product (bonds with coupons e.g.).
(though rarely encountered in practice in the corporate bonds market.)

Remark 6.2. *Valuation of Bonds in a default-free framework.*

We denote by $B(0, T)$ (resp. $B(t, T)$) the price at 0 (resp. t) of a ZCB paying 1 at $T > 0$

(i) *If the interest rate is deterministic, we have*

$$B(0, T) = e^{-\int_0^T r(t)dt} ,$$

with the inf. compound. convention.

(ii) *If the interest rate is stochastic, we have*

$$B(0, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r_t dt} \right] ,$$

where $\mathbb{Q}$ is the risk neutral probability (pricing measure) given by the market.

Remark 6.3. 1. *In the previous assets, Credit Risk is not isolated.*

2. *Loans are placed in the Banking Book and hold until maturity.*

$\hookrightarrow$ *the main component of credit risk is default risk*

3. *Bonds may be held for shorter period of time*

$\hookrightarrow$ *placed in the Trading Book*

$\hookrightarrow$ *subject to default risk and to market risk due to credit quality change of the issuers.*

6.3 Credit Derivatives

Definition 6.2. *A credit derivative is a financial contract whose payoff depends on the occurrence (or non-occurrence) of a specified credit event affecting a predefined reference entity (or a basket of reference entities).*

↔ The credit event—such as default, bankruptcy, or restructuring—is defined contractually and typically follows standard market definitions (e.g. those published by ISDA).

↔ The contract references one or more reference obligations (debt instruments issued by the reference entity), but does not necessarily require ownership of those obligations.

↔ If a credit event occurs during the life of the contract, a contingent payment is triggered, in which the protection seller compensates the protection buyer according to the settlement terms of the contract

key terms:

- *Reference entity/reference credit*: One (or several) issuer(s) whose defaults trigger the credit event. This can be one or several (a basket structure) defaultable issuers.

- *Reference obligations/reference credit asset*: A set of assets issued by reference credit. They are needed for determination of credit event and for calculation of recovery rate which is used to calculate the default payment. Loans and liquidly traded bonds are common choice of reference credit assets.
- *Credit Event*: A precise defined default event. Possible definitions include: bankruptcy, failure to pay, obligation default, ratings downgrade below given threshold (only for ratings-triggered credit derivatives), and change in the credit spread (only for credit spread-triggered credit derivatives).
- *Default payment*: The payment which has to be made if a credit event has happened.
- Credit derivatives are OTC derivatives securities that are linked to a credit-sensitive asset (corporate and sovereign bonds, portfolio of bonds, etc.) as the underlying asset.
 - ↔ They can be used to transfer the credit risk from one counterparty to another and to control credit risk exposure.
 - ↔ Credit derivatives include credit default swaps (CDS), first-to-default swaps, basket default swaps, and collateralised debt obligations (CDO).

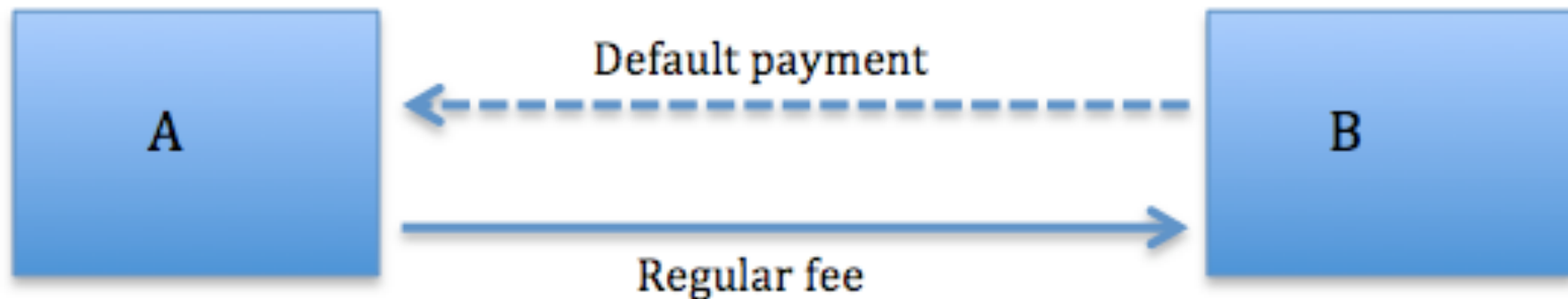
Credit Default Swap (CDS) It is an agreement between two counterparties:

a protection buyer (A) and a protection seller (B). This protection relates to a third party (C).

The setting is:

(A) pays a periodic fee (CDS spread) to (B) until the maturity of the CDS **or** the occurrence of a reference *credit event* of (C) (payment default, rating downgrade, etc.), whichever comes first.

In return (B) promises to pay A some specified payment (*default payment*) if the reference *credit event* for (C) occurs before the maturity of the CDS.



To identify a CDS, one needs to specify:

- the reference obligor (C) and its reference assets;
- the definition of the *credit event*: bankruptcy, failure to pay, obligation default, rating downgrade etc. ;
- the *notional* of the CDS;
- the *CDS spread* (fees rate) and the *frequency of payment* (the day count convention also);
- the *default payment*: it can be physical or cash settled.

↪ physical settlement, the protection seller pays the buyer the notional, and in return takes delivery of a debt obligation (the reference assets).

↪ cash settlement, the protection seller pays the buyer the difference between par value and the market price of the reference assets.

- the start date and *maturity date*.

Remark 6.4. *Credit default swaps are often used to manage the risk of default that arises from holding debt. A bank, for example, may hedge its risk that a borrower may default on a loan by entering into a CDS contract as the buyer of protection. If the loan goes into default, the proceeds from the CDS contract cancel out the losses on the underlying debt.*

Example 6.5. *(Single name CDS)*

- *Reference entity: BNP Paribas*
- *Maturity: 5 years*
- *CDS spread: 150 bps per year*
- *Premium payment: quarterly*
- *Assumed recovery rate: 40%*

Company (A) owns a €10m bond issued by BNPP. It buys from firm (B) the above CDS.

Why? What are the cash flows?

Company (*A*) buys CDS protection to hedge default risk, thus CDS acts as insurance against credit losses.

Premium leg (no default):

$$\text{Annual premium} = 1.50\% \times 10,000,000 = 150,000$$

$$\text{Quarterly payment} = 37,500$$

Protection leg (say default at year 2):

$$\text{Loss} = (1 - 40\%) \times 10,000,000 = 6,000,000$$

$$\text{CDS payout} = 6,000,000$$

Quick overview of CDS

- They have been introduced in the early 1990s. In 1999, the ISDA standardized the documentation for credit default swaps (after LTCM collapsed).
- At first, banks were the dominant players in the market, as CDS were primarily used to hedge risk in connection with its lending activities.
- Banks also saw an opportunity to free up regulatory capital.
- Then, hedge funds became major investors in CDS, seeking trading opportunities and speculating on debt.
- The size of the market had an exponential growth until the GFC then outstanding CDS notionals fell sharply since 2007 peak. (see graph below).
- CDS spreads are widely used as credit risk signals (see pricing section below).

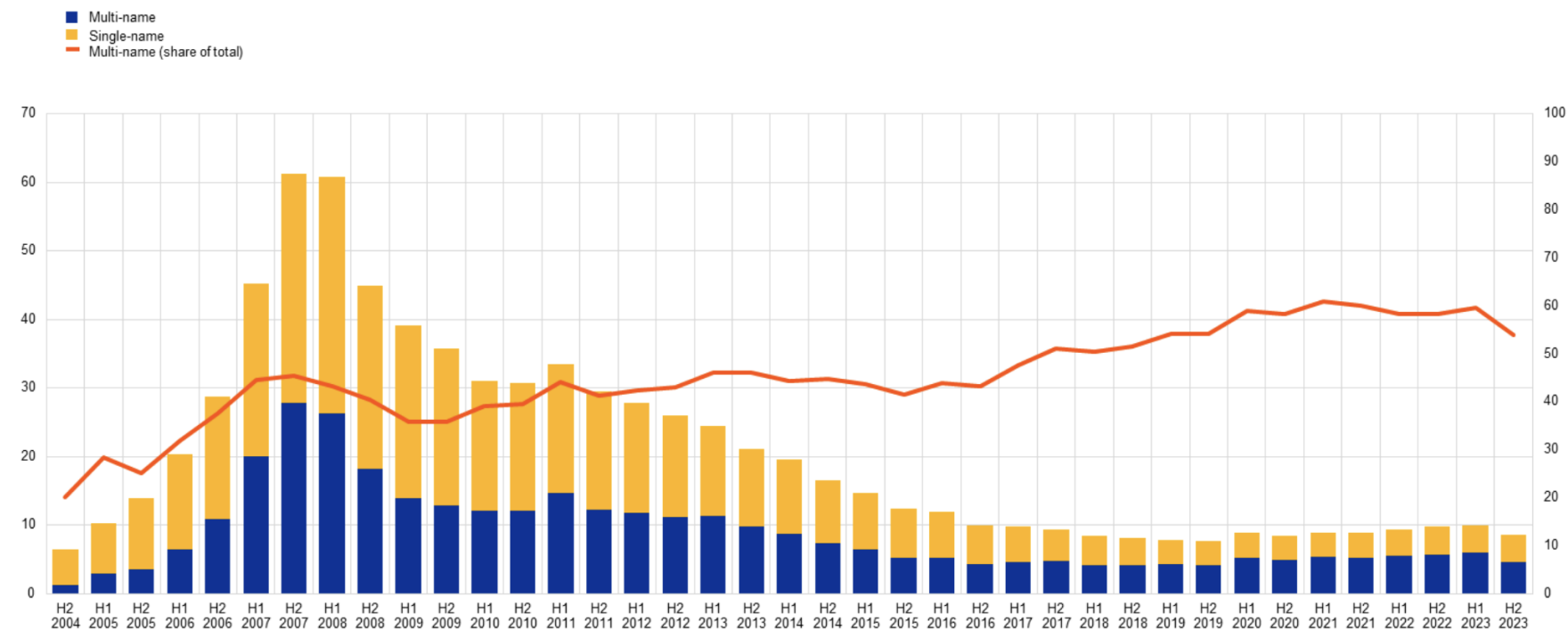
- But the observed CDS spread can reflect limited trading information (low volume, infrequent, clustered a small set of “benchmark” reference obligations per issuer)
- European Systemic Risk Board (ESRB) reports ⁹ that they may cause systemic risk through feedback loop in stress condition e.g.:

bad event → CDS spreads widens → funding conditions tighten & bonds prices fall by reducing exposure (automatic response from risk management) → CDS spreads widens.

⁹https://www.esrb.europa.eu/pub/pdf/reports/esrb.report250411_creditdefaultswaps.en.pdf

Gross notional amounts outstanding for the global CDS market

(USD trillions)



Sources: BIS over-the-counter derivatives statistics and European Systemic Risk Board calculations.

6.4 Credit Rating

- Credit rating is an assessment of relative credit risk of an issuer or a debt instrument:
 - ↪ forward-looking: it reflects expected ability to meet obligations under plausible economic condition
 - ↪ it provides a ranking through the use of letters (generally):
 - AAA, AA, A, BBB, BB, B, CCC, and D with AAA the best quality and D in default.
 - Any bonds rated BBB or above are called investment bonds
 - Those rated BB or below are speculative (junk, high yield) bonds, which have much higher credit risks than those of investment bonds.
- Rating are provided by credit rating agencies (CRA), which assess the creditworthiness of borrowers and debt instruments.
- The global rating market is dominated by three agencies: Standard & Poor's Global Ratings, Moody's Investors Service and Fitch Ratings. (they account for more than 90% of all issued credit ratings worldwide.)
- Ratings are based on expert judgment then various numbers can be attached to them.

Example 6.6. *Probability of Default by rating class*

	<i>1 year</i>	<i>2 years</i>	<i>3 years</i>	<i>4 years</i>	<i>...</i>
<i>AAA</i>	0.00	0.00	0.02	0.05	<i>...</i>
<i>AA</i>	0.01	0.03	0.07	0.14	<i>...</i>
<i>A</i>	0.03	0.07	0.12	0.25	<i>...</i>
<i>BBB</i>	0.29	0.60	1.00	1.80	<i>...</i>
<i>BB</i>	2.30	3.80	6.50	9.00	<i>...</i>
<i>B</i>	5.29	12.00	16.10	19.40	<i>...</i>
<i>CCC</i>	45.56	56.80	62.10	68.50	<i>...</i>
<i>D</i>	100				

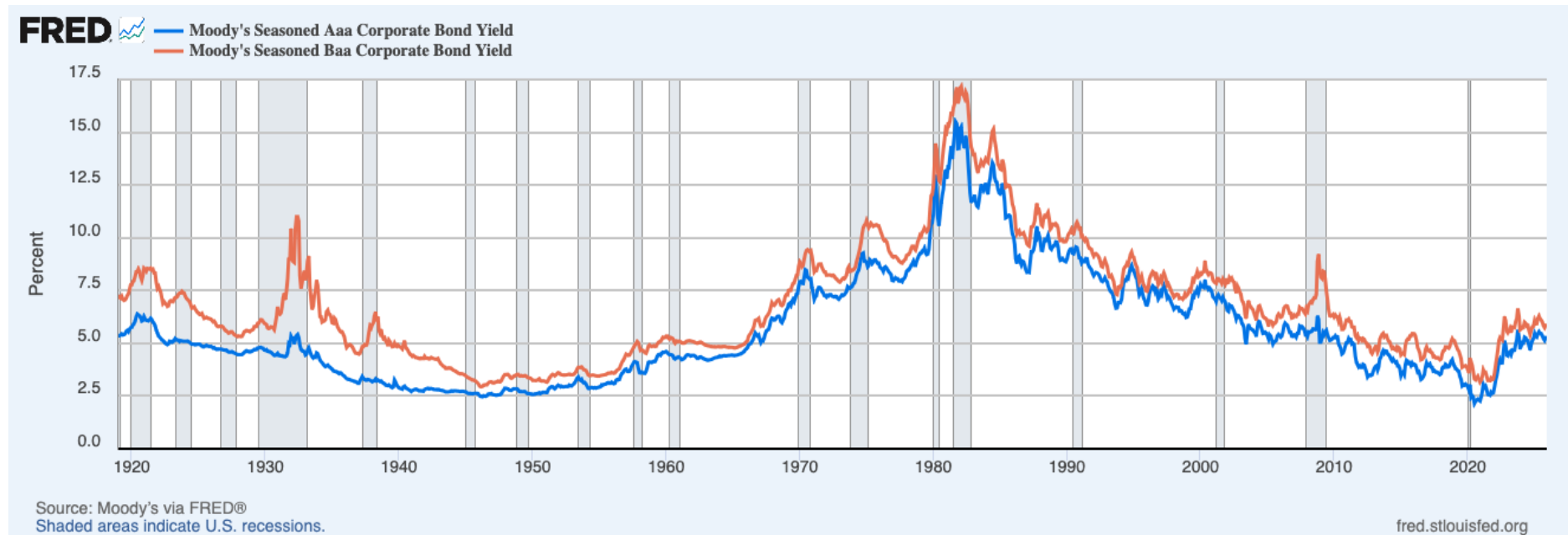
- Note: historical average frequencies do not reflect newly available market information nor distinctions among firms of the same rating.

Table 4: Moody's Average Ratings Transition Frequency Matrix (1980–2000)

Initial rating	Rating at year end (percent)							
	Aaa	Aa	A	Baa	Ba	B	Caa–C	Default
Aaa	89.14	9.78	1.06	0.00	0.03	0.00	0.00	0.00
Aa	1.14	89.13	9.25	0.32	0.11	0.01	0.00	0.03
A	0.06	2.97	90.28	5.81	0.69	0.18	0.01	0.01
Baa	0.06	0.36	7.01	85.47	5.82	1.02	0.08	0.17
Ba	0.03	0.07	0.59	5.96	82.41	8.93	0.58	1.44
B	0.01	0.04	0.22	0.61	6.43	82.44	3.29	6.96
Caa–C	0.00	0.00	0.00	0.95	2.85	6.15	62.36	27.68

The rating agencies publish also historical average transition frequencies among credit ratings and into default.

Example of rating impact on bonds yield (US long-term bonds)



note: “historical illustration”, not used for pricing. source: <https://fred.stlouisfed.org/>

6.5 How credit risk is embedded in prices?

In the following, we work with a risk neutral probability $\mathbb{Q}$ (chosen by the market).

6.5.1 Risk neutral probability of Default

Default-free ZCB The risk-free rate is a stochastic process denoted $(r_t)_{t \geq T}$.

A risk-free ZCB with maturity T has price:

$$B(0, T) := \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r_s ds} \right]$$

Defaultable ZCB The payoff at maturity of this ZCB is

$$\mathbf{1}_{\{\tau > T\}} + \mathcal{R} \mathbf{1}_{\{\tau \leq T\}}$$

$\mathcal{R} \in [0, 1)$ is the recovery, assumed for ZCB to be paid at maturity.

- We assume that $\mathcal{R}$ is deterministic and τ is independent of $(r_t)_{t \geq 0}$

Pricing in the case of zero recovery:

We obtain that the price $\bar{B}$ at 0 of the defaultable ZCB is given by

$$\bar{B}(0, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r_s ds} \mathbf{1}_{\{\tau > T\}} \right] = B(0, T) \mathbb{Q}(\tau > T) .$$

Definition 6.3.

- The quantity $\mathbb{Q}(\tau > T)$ is the *implied survival probability*
- The quantity $\mathbb{Q}(\tau \leq T) = 1 - \mathbb{Q}(\tau > T)$ is the *implied default probability* over the period $[0, T]$.
- We denote by $\text{PD}^* = \mathbb{Q}(\tau \leq 1)$ the one year default probability implied by the market.
- we denote by $f^*(\cdot)$, the p.d.f. of τ under $\mathbb{Q}$ (assumed to exist).
- Given $0 < \dots < t_i < t_{i+1} < \dots < T$, the *implied probability of default* on $(t_i, t_{i+1}]$ is

$$\mathbb{Q}(t_i < \tau \leq t_{i+1})$$

Let us remark:

$$\mathbb{Q}(\tau \leq T) = \mathbb{Q}(\tau = 0) + \sum_{i=0}^{n-1} \mathbb{Q}(t_i < \tau \leq t_{i+1}),$$

- The *implied conditional survival probability* over the period t_i, t_{i+1} is given by

$$\mathbb{Q}(\tau > t_{i+1} | \tau > t_i) = \frac{\mathbb{Q}(\tau > t_{i+1})}{\mathbb{Q}(\tau > t_i)}$$

and *implied conditional PD* on $[t_i, t_{i+1}]$ is

$$\mathbb{Q}(\tau \leq t_{i+1} | \tau > t_i) = \frac{\mathbb{Q}(t_i \leq \tau \leq t_{i+1})}{\mathbb{Q}(\tau > t_i)}$$

Credit spread: The quantity $s(0, T)$ such that

$$s(0, T) := -\frac{1}{T} \ln \left(\frac{\bar{B}(0, T)}{B(0, T)} \right) .$$

- If we denote by $\bar{y}(0, T)$ the yield to maturity of the defaultable ZCB and $y(0, T)$ the yield to maturity of the ZCB without default risk, we have

$$\bar{y}(0, T) = y(0, T) + s(0, T)$$

- In the case of zero recovery, $\mathbb{Q}(\tau > T) = e^{-s(0, T)T}$,
- If $T = 1$ year, then $s(0, 1) \approx \text{PD}^*$, if the credit spread is relatively small.

Example 6.7. *Credit spreads term structure*

<i>maturity</i> (y)	0.5	1	3	5	7	10
<i>ZCB</i>	0.9667	0.9272	0.7883	0.6605	0.5442	0.3962

The risk-free interest rate is $r = 5\%$.

(i) Find the yield to maturity of the ZCB.

(ii) Draw the credit spreads term structure.

Computed Yields to Maturity & Spreads

T (years)	$\bar{B}(0, T)$	$\bar{y}(0, T)$	$s(0, T) = \bar{y}(T) - 5\%$
0.5	0.9667	6.773%	1.773%
1	0.9272	7.559%	2.559%
3	0.7883	7.929%	2.929%
5	0.6605	8.295%	3.295%
7	0.5442	8.692%	3.692%
10	0.3962	9.258%	4.258%

Pricing in the non-zero recovery case: Let us now assume that $\mathcal{R} \neq 0$ (but still deterministic).

- The payoff at maturity of the defaultable ZCB is

$$\mathbf{1}_{\{\tau > T\}} + \mathcal{R}\mathbf{1}_{\{\tau \leq T\}} = 1 - (1 - \mathcal{R})\mathbf{1}_{\{\tau \leq T\}}$$

(we assume recovery at maturity)

- Then, we have:

$$\bar{B}(0, T) = B(0, T) \left(1 - (1 - \mathcal{R})\mathbb{Q}(\tau \leq T) \right) \text{ and } s(0, T) = -\frac{1}{T} \ln \left(1 - (1 - \mathcal{R})\mathbb{Q}(\tau \leq T) \right)$$

$\hookrightarrow$ The credit spread is smaller when there is non zero recovery.

- If $T = 1$, then $s(0, 1) \approx (1 - \mathcal{R})\text{PD}^*$, which corresponds to a one year expected loss (risk neutral).
- From market prices, you can get $(1 - \mathcal{R})\text{PD}^*$ and then assuming a value for $\mathcal{R}$, PD^* these market PD are higher than the historical PD: there is a risk premium embedded.

6.5.2 Pricing defaultable fixed coupon bond

- Defaultable bond:

- Notional $\mathcal{N}$

- Coupon C paid at regular date $(t_n)_{1 \leq n \leq N}$ up to maturity $T =: t_N$.

$\hookrightarrow$ Thus: at time $0 < t_n < T$: $F_{t_n} = C$, for $1 \leq n \leq N$ and at time T : $F_T = C + \mathcal{N}$

- In the case of default, some coupons and the notional are not paid.

$\hookrightarrow$ the buyer of the bond recovers only part of the notional after the default time.

- The stochastic discounted value of the bond is then

$$G = \sum_{i=n}^N F_{t_n} e^{-\int_0^{t_n} r_s ds} \mathbf{1}_{\{\tau > t_n\}} + \mathcal{R} N e^{-\int_0^T r_s ds} \mathbf{1}_{\{\tau \leq T\}} \quad (6.1)$$

the market price of such claim is given by

$$p_0 = \mathbb{E}^{\mathbb{Q}}[G] = \sum_{n=1}^N C B(0, t_n) \mathbb{Q}(\tau > t_n) + \mathcal{N} B(0, T) \mathbb{Q}(\tau > T) + \mathcal{R} \mathcal{N} \int_0^T B(0, t) f^*(t) dt .$$

6.5.3 Pricing CDS

- Fixed leg (premium leg): the fees over the period t_{n-1}, t_n are $\delta(t_n - t_{n-1})$ if no default has occurred before t_n i.e.

$$\mathcal{N}\delta(t_n - t_{n-1})\mathbf{1}_{\{\tau > t_n\}}$$

- Stochastic discounted value of the fixed leg:

$$G_{FixL} = \mathcal{N} \sum_{n=1}^N \delta(t_n - t_{n-1}) e^{-\int_0^{t_n} r_s ds} \mathbf{1}_{\{\tau > t_n\}}$$

-The market price of the Fixed leg is

$$p_{FixL}(0) = \mathbb{E}^{\mathbb{Q}} [G_{FixL}] = \mathcal{N} \sum_{n=1}^N \delta(t_n - t_{n-1}) B(0, t_n) \mathbb{Q}(\tau > t_n) .$$

- Floating Leg (protection leg): If a default occurs before T , then $\mathcal{N}(1 - \mathcal{R})$ is paid after the default, its stochastic discounted value is

$$G_{FloL} = \mathcal{N}(1 - \mathcal{R}) \mathbf{1}_{\{\tau \leq T\}} e^{-\int_0^\tau r_s ds}$$

its market price is

$$p_{FloL}(0) = \mathbb{E}^{\mathbb{Q}} [FloL] = \mathcal{N}(1 - \mathcal{R}) \int_0^T B(0, t) f^*(t) dt$$

- In particular the CDS spread is given such that $p_{FloL}(0) = p_{FixL}(0)$ and thus

6.6 Credit risk modelling

- Key point is to model the default time (note also other parameters may be considered as e.g. recovery)
- Two main Types of model that are complementary:
 1. Intensity based model or reduced form model: default as the arrival of an exogenous random time with an intensity (hazard rate) calibrated from market prices. $\Leftrightarrow$ There is no economic explanation of default.
 2. Structural model (see next section!) A structural credit risk model explains default as an endogenous event occurring when a firm's asset value falls below a debt threshold

6.7 A glimpse of reduced-form model

- We just consider the simple case of

$$\tau \sim \mathcal{Exp}(\lambda).$$

- The first jump of a Poisson process. For stock model in credit risk, one relies on jump processes [7].

$\hookrightarrow$ See the course on Credit Risk!

- In reduced-form model, the key variable is the (risk neutral) hazard rate

$$h(u) = \lim_{\Delta u \rightarrow 0} \frac{1}{\Delta u} \mathbb{Q}(u < \tau \leq u + \Delta u \mid \tau > u)$$

- In our simple setting, $h(u) = \lambda$, for all $u \geq 0$. The survival probability is $\mathbb{Q}(\tau > T) = e^{-\lambda T}$.
- In our simple example: if one assumes *zero recovery* for the ZCB, one obtains a flat term structure for credit spread, namely $s(0, T) = \lambda$.

CDS spread:

- Using (6.2), namely

$$\delta = \frac{(1 - \mathcal{R}) \int_0^T B(0, t) f^*(t) dt}{\sum_{n=1}^N (t_n - t_{n-1}) B(0, t_n) \mathbb{Q}(\tau > t_n)}$$

- Approximate the premium leg by its continuous version so that

$$p_{FixL}(0) \approx \mathcal{N} \delta \int_0^T B(0, t) \mathbb{Q}(\tau > t) dt .$$

- We get that $\delta \approx (1 - \mathcal{R}) \text{PD}^*$

- The CDS spread is approximatively the one-year expected loss (risk-neutral). It can indeed be interpreted as a credit risk measure of the reference entity!

7 Structural Models

- The structural approach models credit risk by referring directly to economic fundamentals such as the capital structure of a company.
- Two key ingredients of this approach are the *total value of the firm's asset* and the *default triggering barrier*.
- We present the seminal Merton's model [17] and discuss rapidly some extensions, in particular Black & Cox model [6].

7.1 Merton's Model

Merton's model [17] is based on the Black and Scholes model for stock price. It relies on several assumptions:

1. No transaction costs, no tax. Assets are divisible and traded continuously.
2. There are sufficiently many investors with comparable level of wealth such that they can buy and sell as much as they want at a given market price.
3. There is a constant risk free interest rate r . (same rate for borrowing and lending money). There is no restriction on borrowing and lending money.
4. The total asset value of the firm, V , follows a geometric Brownian motion under the risk neutral probability denoted by $\mathbb{Q}$ (W is a standard Brownian motion under $\mathbb{Q}$)

$$dV_t = V_t(rdt + \sigma_V dW_t) \tag{7.1}$$

where σ_V is a constant.

5. The firm is completely described by its balance sheet:

<i>Asset</i>	<i>Liability</i>
$V(asset\ value)$	$\bar{B}\ (Debt)$
	$S\ (Equity)$

The total value of asset is financed by equity S and one representative zero-coupon non-callable debt contract $\bar{B}$ maturing at time T and paying $\bar{D}$ (notional + interests).

$\hookrightarrow$ Accounting identity of the balance sheets: for all $0 \leq t \leq T$,

$$V_t = S_t + \bar{B}(t, T).$$

6. There are neither cash flow payouts, nor issues of any type of security during the life of the debt contract, which implies that the default can only occur at maturity.

$\hookrightarrow$ Default time $\tau = T\mathbf{1}_{\{V_T < \bar{D}\}} + \infty\mathbf{1}_{\{V_T \geq \bar{D}\}}$

7. The absolute priority rule cannot be violated, i.e., shareholders obtain a positive payoff only in the case that bondholders are perfectly reimbursed.

$\hookrightarrow$ the *owner of the bond* will receive at maturity

$$X_T = V_T \wedge \bar{D} = \bar{D} - [\bar{D} - V_T]^+$$

This means that the bond's holder is long a risk free ZCB and short a put on the firm's asset value of strike $\bar{D}$.

$\hookrightarrow$ *value of the equity* at maturity is

$$S_T = [V_T - \bar{D}]_+$$

This payoff is a call on the firm's assets value with strike $\bar{D}$.

Pricing Use the BS formula to derive a no-arbitrage price of the risky bond and the equity at any time $t \in [0, T]$. (Φ denotes the cumulative distribution function of the standard normal law)

$\hookrightarrow$ *The equity price* is given by

$$S_t := \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} [V_T - \bar{D}]_+ \mid \mathcal{F}_t \right] = V_t \Phi(d_+) - \bar{D} e^{-r(T-t)} \Phi(d_-)$$

where

$$d_{\pm} = \frac{-\ln(\delta_t) \pm \frac{1}{2}\sigma_V^2(T-t)}{\sigma_V\sqrt{T-t}} \quad \text{and} \quad \delta_t = \frac{\bar{D}e^{-r(T-t)}}{V_t} \quad (\text{quasi debt-to-asset ratio})$$

In particular, $S_t = u(t, V_t)$ where u is a solves the Black-Scholes PDE.

$\hookrightarrow$ The Bond price is given by

$$\bar{B}(t, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} X_T \mid \mathcal{F}_t \right] = V_t - S_t = V_t \Phi(-d_+) + \bar{D} e^{-r(T-t)} \Phi(d_-).$$

It is also a function of (t, V_t) .

$\hookrightarrow$ The **risk-neutral** probability of default at time t is given by

$$\mathbb{Q}(\tau \leq T \mid \mathcal{F}_t) = \mathbb{Q}(\tau = T \mid \mathcal{F}_t) = \Phi(-d_-)$$

$\hookrightarrow$ *The yield to maturity* at time t is given by

$$\bar{y}(t, T) = -\frac{1}{T-t} \ln\left(\frac{\bar{B}(t, T)}{\bar{D}}\right) \text{ i.e. } \bar{B}(t, T) = \bar{D}e^{-(T-t)\bar{y}(t, T)}$$

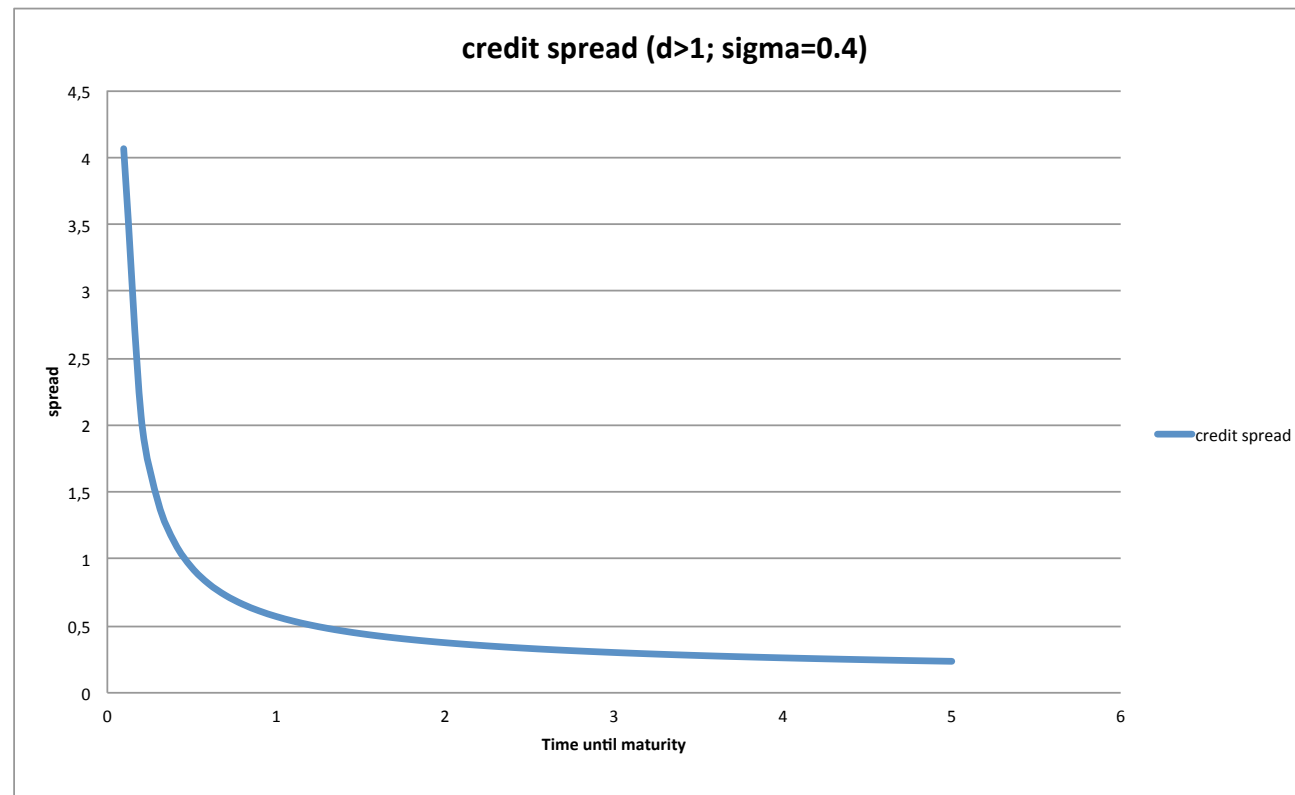
$\hookrightarrow$ *The credit spread* is given by

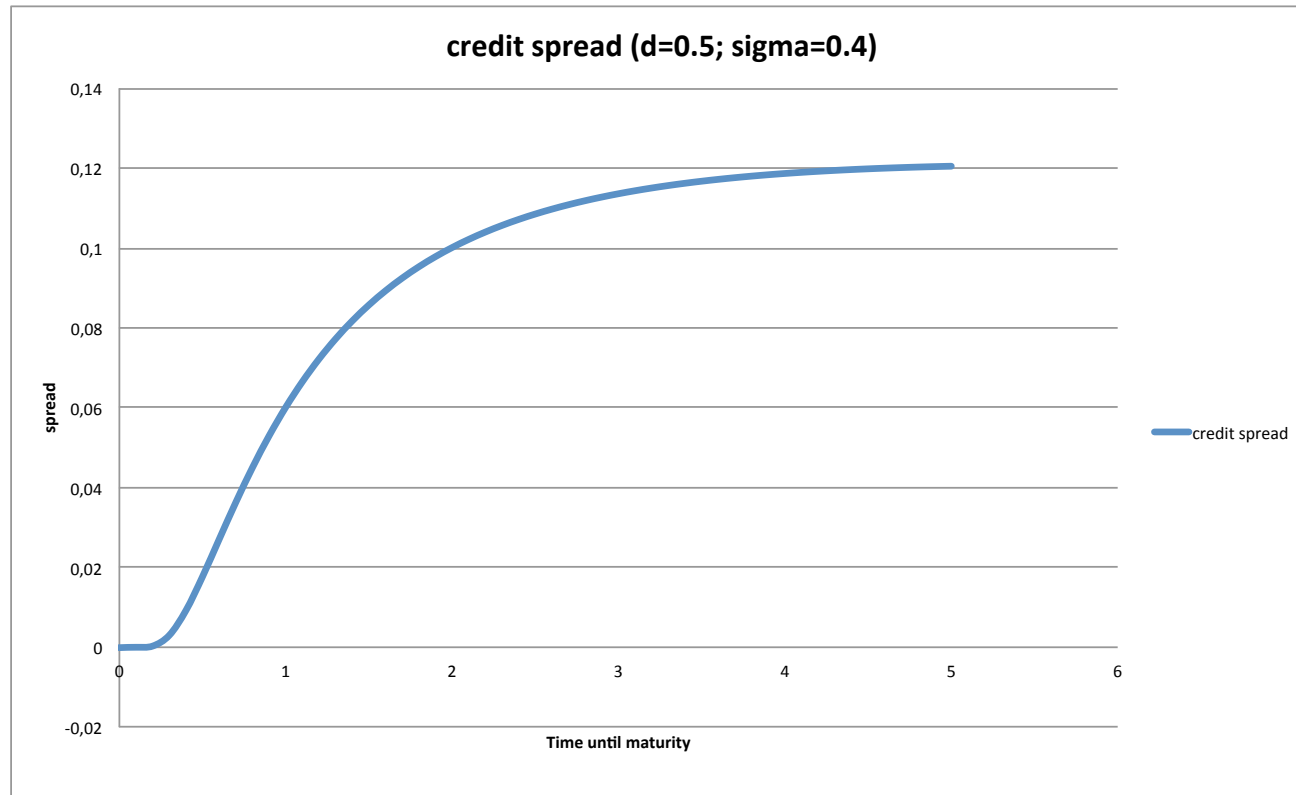
$$(i) \ s(t, T) = \bar{y}(t, T) - r > 0$$

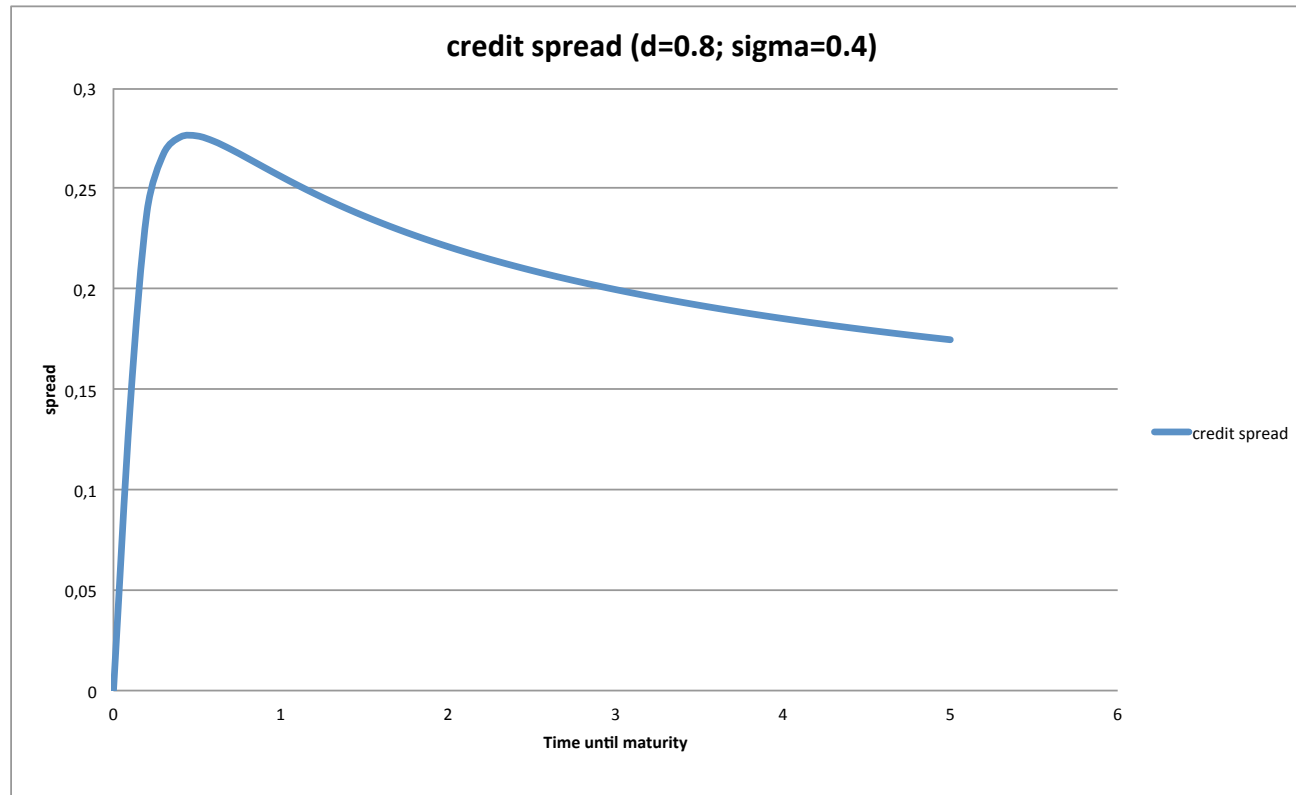
and satisfies

$$(ii) \ \lim_{t \uparrow T} s(t, T) = \begin{cases} \infty & \text{if } \delta_T > 1 \\ 0 & \text{if } \delta_T < 1 \end{cases}$$

Some graphs for various value of δ and σ :







- **Calibration of firm asset value:** Note that for listed companies the observable are S_0 and σ_S (from option or historical stat.).
- Most often V_0 and σ_V are not observed but Ito's Lemma applied to $t \mapsto S_t = u(t, V_t)$ implies:

$$\sigma_S = \frac{1}{S} \frac{\partial S}{\partial V} V \sigma_V \quad (7.3)$$

leading to, at $t = 0$,

$$\sigma_V = \frac{S_0}{\Phi(d_+) V_0} \sigma_S. \quad (7.4)$$

- Combining the previous equation with the formula for the equity S_0 we obtain the system:

$$S_0 = V_0 \Phi(d_+(0, V_0, \sigma_V)) - \bar{D} e^{-rT} \Phi(d_-(0, V_0, \sigma_V)) \quad (7.5)$$

$$\sigma_V = \frac{S_0}{\Phi(d_+) V_0} \sigma_S. \quad (7.6)$$

Using a numerical procedure, one may solve the above system for the values for (V_0, σ_V) .

$\hookrightarrow$ This is a typical application of Merton's model (see KMV model below).

7.2 Black and Cox model

- Merton's model has a restrictive and unrealistic feature: the default can only happen at maturity of the debt.
- Black and Cox [6] extend Merton's model in several directions by taking into account more specific features of debt contracts.
- They consider in particular, *safety covenants* which allow the bondholders to force the firm to bankruptcy if its value falls below some deterministic level (time dependent).
- The default of the firm is thus triggered by this level: default can happen anytime between 0 and T .

The setup is the same as in Merton's model, the value of the firm satisfies the Black Scholes SDE (7.1) (see also the other assumptions of BS model).

- BUT now the default triggering barrier is given by:

$$\ell_t = Ke^{-\gamma(T-t)}, \quad K > 0.$$

assuming that $\ell_t \leq \bar{D}e^{-r(T-t)}, \forall t$.

- The *default time* is defined by

$$\tau := \bar{\tau} \wedge \hat{\tau}$$

where $\hat{\tau}$ is Merton's default time: $\hat{\tau} := T\mathbf{1}_{\{V_T < \bar{D}\}} + \infty\mathbf{1}_{\{V_T \geq \bar{D}\}}$

and $\bar{\tau} = \inf\{t \geq 0 \mid V_t \leq \ell_t\}$.

The pricing is done similarly to barrier option (down and out call for equity).

7.3 Comments

Various extensions were considered see [18]:

- the bankruptcy barrier is endogenously determined from the optimal behaviour of the debtors [15]
- problem of low credits spread for short maturity: jumps in the dynamics of the assets value of the firm [21], imperfect information on the value of the firm [9] ...
- The CreditGrades ¹⁰ software [11] is based on Black & Cox model.

It was used by credit risk analysts to value credit risk instruments like credit default swaps (CDS) and to supplement other credit assessments. (not really used anymore)

¹⁰<https://www.msci.com/documents/10199/dd31bcce-6fe3-47b7-9fb7-10c4c8f750ba>

8 Credit risk assessment

- Various risk can be taken into account:
 - Spread Risk: credit-related market risk arising from changes (volatility) in credit spreads (e.g. CDS or bond spreads).
 - "Migration" risk: risk due to rating changes.
 - Default risk: the risk of counterparty default, including uncertainty on recovery.
- There are two possible views:
 1. Trading-book credit portfolio:
 - FRTB does take into account spread risk and jump to default risk $\implies$ Regulatory capital
 - A “good” Economic Capital model will:

1. take into account *both* rating migration and spread volatility movement
2. capture correlation for portfolio, model stochastic recovery etc.

↪ require sophisticated model of credit risk.

2. In the banking book:

- The regulatory Basel IRB formula works in default mode (taking into account only the default, see below).
- A “good” Economic Capital model will:
 1. still work in default mode on a one-year horizon
 2. relax strong assumptions of the IRB formula (use multi-factor models, stochastic recovery etc.)

- For the banking book, [Credit Risk is evaluated over a one-year horizon](#)¹¹ Some arguments:
 - The firm's main accounting cycle,
 - A reasonable period to renew any capital depleted through losses,
 - A reasonable period over which actions can be taken to mitigate losses,
 - Credit reviews and borrowers evaluation are usually performed annually.

¹¹It contrasts really from Market Risk where the horizon is a couple of days.

8.1 Loss in MtM approach

- For debt securities (bonds, credit derivatives), it takes into account spread risk & rating changes.
- For an individual asset on time horizon T :
 - at $t = 0$ the asset is in rating $\mathbf{r}_0 = i$, its value $V^{(i)_0}$ (observed on the market or computed from a model)
 - draw the rating $\mathbf{r}_T$ at time T according to a rating transition matrix (p_{ij}) namely $\mathbb{P}(\mathbf{r}_T = j | \mathbf{r}_0 = i) = p_{ij}$
 - evaluate asset at time T in rating $\mathbf{r}_T$: $V_T^{(\mathbf{r}_T)}$
 - the loss is then

$$L = V_0^{(i)} - V_T^{(\mathbf{r}_T)} \tag{8.1}$$

↪ Computed using Monte Carlo simulation and then risk measures can be estimated.

- Remark: it is not so easy to construct the transition matrix.

8.1.1 Creditmetrics

- We discuss the creditmetrics approach for a single asset and the portfolio model: this legacy model works only in "migration mode". (no spread vol taken into account, only rating changes)
 1. this model was primarily designed for banking-book-type credit portfolios. It was done 10 year (or so) before the crisis (sparse data on spread, CDS spread etc. mostly rating where dominating)
 2. this has inspired new EC model but it cannot be used anymore as a standalone or realistic EC model (too simplistic)
- The horizon is set to $T = 1$ year.

For a single asset: *Example* loss distribution of a BBB bond (stand alone)

Features of this bond: maturity = 5Y, annual coupon=6%, (remark: one should also report the level of protection – seniority– of the bond)

- Cashflows:

- in any rating but default: $\mathcal{N} \times 6\%$ at year 1, 2, 3, 4 and $\mathcal{N} \times 106\%$ at year 5.

- if default: $\mathcal{R} \times \mathcal{N}$ ($\mathcal{R}$ follows a beta distribution)

- *Loss computation:*

Step one: Specify the transaction matrix (historical Moody's say)

Step two: Specify the forward pricing model.

Assume a *static* spread curve per rating class. (note: This implicitly assumes no spread volatility and no rate uncertainty.)

- The spot zero curve $\rightarrow$ the current spot value of the bond.

$$V_0^{(\mathbf{r}_0)} = \sum_{n=1}^5 \mathcal{N} \times 6\% \times \bar{B}^{\mathbf{r}_0}(0, n) + \bar{B}^{\mathbf{r}_0}(0, 5) \mathcal{N}$$

where $\bar{B}^{\mathbf{r}_0}(0, n)$ is the discount factor associated to the bond in rating $\mathbf{r}_0$

- Since rates are static (i.e., today's term structure fully determines future discounting), then pricing “in one year” is just forward pricing using today's curve.
- - If the rating is $\mathbf{r}_1$ (not default)

$$V_1^{(\mathbf{r}_1)} = \sum_{n=1}^5 \mathcal{N} \times 6\% \times \frac{\bar{B}^{\mathbf{r}_1}(0, n)}{\bar{B}^{\mathbf{r}_1}(0, 1)} + \frac{\bar{B}^{\mathbf{r}_1}(0, 5)}{\bar{B}^{\mathbf{r}_1}(0, 1)} \mathcal{N}$$

(namely use the one year forward curve)

- If default, $V_1^D = \mathcal{R} \times \mathcal{N}$

- The loss that has to be simulated (since the rating is random at $t = 1$ and possibly the recovery):

$$L = V_0^{(\mathfrak{r}_0)} - V_1^{(\mathfrak{r}_1)}$$

- One can then compute VaR.

The case of a portfolio

Example: loss distribution of portfolio of corporate bonds or loans

A new element is now mandatory: the dependence between rating changes of all the underlying bonds/loans.

Facts:

- “Correlations” tend to be higher for firms within the same industry or in the same region.
- “Correlations” vary with the relative state of the economy in the business cycle.

During a recession, most of the assets of the obligors will decline in value and quality, and the likelihood of multiple defaults increases substantially. The contrary happens when the economy is performing well.

↔ CreditMetrics deals with the first point, but not the second one: a “correlation model” is built.

Assumption: There is an underlying structural model. The correlation between equity returns $\approx$ the correlation between asset returns.

The asset value V_i of any firm i belongs to some “bucket”, given by the current rating of i .

It is [simpler to think in terms of normal variables](#). We associate a normal r.v. Z_i to every firm i . We can assimilate i 's asset value and Z_i . (see next section for a justification)

$\hookrightarrow$ Correlation between the (Z_i) will lead to correlation between the rating migration (as follows):

The initial rating of i is $\mathbf{r}_0^i$ (known) and its rating at T is $\mathbf{r}_T^i$ (unknown, i.e. random).

Value of Z_i are mapped to the rating migration as follows (All these thresholds depend on the initial rating $\mathbf{r}_0^i$):

$$\left\{ \begin{array}{l} P(\mathbf{r}_T^i = AAA|\mathbf{r}_0^i) = P(Z_i > z_{AAA}), \\ P(\mathbf{r}_T^i = AA|\mathbf{r}_0^i) = P(Z_i > z_{AA}) - P(Z_i > z_{AAA}), \\ P(\mathbf{r}_T^i = A|\mathbf{r}_0^i) = P(Z_i > z_A) - P(Z_i > z_{AA}), \\ P(\mathbf{r}_T^i = BBB|\mathbf{r}_0^i) = P(Z_i > z_{BBB}) - P(Z_i > z_A), \\ P(\mathbf{r}_T^i = BB|\mathbf{r}_0^i) = P(Z_i > z_{BB}) - P(Z_i > z_{BBB}), \\ P(\mathbf{r}_T^i = B|\mathbf{r}_0^i) = P(Z_i > z_B) - P(Z_i > z_{BB}), \\ P(\mathbf{r}_T^i = CCC|\mathbf{r}_0^i) = P(Z_i > z_{CCC}) - P(Z_i > z_B), \\ P(\mathbf{r}_T^i = D|\mathbf{r}_0^i) = P(Z_i < z_{CCC}). \end{array} \right.$$

Correlating all the gaussian r.v. Z_i , $i = 1, \dots, n$, we are able to simulate jointly all the rating transitions in the underlying portfolio.

In practice, this correlation matrix can be huge, depending on the number of obligors.

Thus, some factor models are built: a grid of industry $\times$ country correlated factors.

Every name is decomposed linearly on the industry $\times$ country factors where it is involved.

Ex: $Z_i = 0.6 \mathcal{F}_{US,chemicals} + 0.2 \mathcal{F}_{Fr,auto} + \sqrt{1 - 0.6^2 - 0.2^2} \varepsilon_i$.

- Value at time T can then be computed for each bonds as done in the single asset case. Aggregation leads to value of the portfolio at time T .

$\hookrightarrow$ then to the overall loss.

8.1.2 Distance to Default

- A structural approach to justify the above view.
- Firm's total assets dynamics: Assumed to be log-normal

$$dV_t = V_t(\mu dt + \sigma_V dW_t) \tag{8.2}$$

(under the physical probability!)

Setting $Z = \frac{W_T}{\sqrt{T}}$, one gets from

$$V_T = V_0 e^{(\mu - \frac{\sigma_V^2}{2})T + \sigma_V \sqrt{T} Z}$$

the expression of Z .

- Having in mind that default happens at T if and only if $V_T \leq V_d$ (V_d is the default threshold – depends on the applications) we get

$$V_T \leq V_d \iff Z \leq -DD$$

where

$$DD := \frac{\log(\frac{V_0}{V_d}) + (\mu - \frac{\sigma_V^2}{2})T}{\sigma_V \sqrt{T}} = \frac{\mathbb{E}[\log(V_T)] - \log(V_d)}{\sigma_V \sqrt{T}} \quad (8.3)$$

- say $T = 1$ year (as it is usually the case) then

$$PD = \mathbb{P}(Z \leq -DD) \quad \text{i.e.} \quad DD = -\Phi^{-1}(PD).$$

- What is done for the extreme rating "default" can be done for the other rating as presented above. One main point in the Creditmetrics procedure is that this various classes are obtained from external synthetic rating system (as S&P or Moody's).

8.1.3 Moody's KMV

- KMV approach is somehow the converse as it computes the DD directly from structural model
- The first step is to obtain V_0 and σ_V via their link with S_0 and σ_S (see Merton's Model section)
- Then the drift of the firm's asset value has to be fixed: it can be set to r plus some premium that has to be estimated.
- V_d is set to the sum of the short term debt and one-half the medium to long term debt. (empirically observed)
- Now the value DD computed with these data can be considered as a score (to build system rating).
- MKV *does not* set $PD = \Phi(-DD)$! Instead it computes an Expected Default Frequency by mapping the DD firm to the frequency of default historically displaying the same DD. So, $EDF = f(DD)$ for some monotone function f .
- This EDF can in turned be mapped to any rating system (S&P, or internal to the bank etc.).
- Note that EDF are more sensitive to economic conditions than S&P ratings.

8.2 Loss in default default mode

- For an horizon T : the loss depends on the Exposure at Default (EAD), the Loss Given Default (LGD) (a percentage) and the timing of default.

- Loss definition:

- For a single asset with default time τ ,

$$L = EAD \times LGD \times \mathbf{1}_{\{\tau \leq T\}} \quad (8.4)$$

- For a portfolio of assets, simply

$$\sum_{n=1}^N L_n \text{ with } L_n = EAD_n \times LGD_n \times \mathbf{1}_{\{\tau_n \leq T\}}. \quad (8.5)$$

with τ_n default time of asset n .

• **EAD = Exposure At Default** = the (monetary) amount that a lender "should" lose if an obligor defaults at some time horizon

- EAD may be related to a single deal, a set of deals (netted or not), a counterparty etc.
- One of the main issues for many types of credit instruments is that a bank's exposure may not be known with certainty, but depends on the occurrence of some future events: when credit is optional (lines of credit), dependence on some market factors...
- All these random factors cannot be easily evaluated, and are partly arbitrary.

↪ For Loans and Bonds (our main focus below), we consider simply the par value as EAD.

- **Loss Given Default** = the percentage of the credit exposure that the lender will lose if the borrower defaults. It is also referred to as loss “severity”.

$$LGD = 1 - \text{Recovery rate.} \quad (8.6)$$

- *Remark:* in view of the LGD definition (8.6), the loss defined above is understood (for a loan) as

$$\text{loss} = \text{par value} - \text{value at default (i.e. recovery) upon default}$$

↪ such models indeed disregard losses due to rating migration

- A high level of uncertainty around future realized LGDs, Some important factors determining LGD are:

1. *Seniority:* the higher the seniority, the lower the LGD. (Derivatives are normally *pari passu* with the senior bondholders.)
2. *Collateral.* Legal difficulties may prevent the lender from accessing the collateral, even if it is in the form of deposits.

3. *Type of borrower* (sovereigns, corporations, individuals) and/or obligation (secured loan, credit card debt, mortgage...). SME tend to show higher LGDs than larger corporates.
4. *industry & jurisdiction/country*
5. *the business cycle*. In a recession, lenders are hit twice: more defaults than usual, and higher severities.

Depending of the type of counterparties and the nature of the instruments, LGDs are evaluated differently:

1. **Market value recovery**: $\mathcal{R}$ = the market value per unit of legal claim amount of defaulted debt, some short time after default (one or a few months) – more a trading-book approach
2. **Settlement value recovery**: $\mathcal{R}$ = value of the default settlement per unit of legal claim, discounted back to the date of default, after subtracting legal/administrative costs. – more a banking-book approach

Seniority Class	Average Rec Rate(%)
First lien bank loan	65.8
Second lien bank loan	29.1
Senior unsecured bank loan	47.8
Senior secured bond	50.8
Senior unsecured bond	36.7
Senior subordinated bond	30.7
Subordinated bond	31.3
Junior subordinated bond	24.7

Table 5: Historical recovery rates as a function of seniority levels (Moody's, 1982-2010).

8.2.1 A measure of credit risk: Expected Loss (EL) – in *default mode*

- EL is the amount that an institution expects to lose on a credit exposure over a given time horizon T .
- Expected loss is not bad luck: it is what you should expect to happen.
- For a single asset (assuming some independence property):

$$\text{Expected Loss}(T) = \text{Probability of Default} \times \text{Expected Exposure at Default} \times \text{Loss Given Default}.$$

- The expected loss for a portfolio $\mathcal{L}$ is the sum of the individual expected losses (L_i)

$$\sum_{n=1}^N \mathbb{E}[L_n]$$

The expectation above is under the **PHYSICAL PROBABILITY**. (This is the risk management view, not pricing view.)

- This expected loss is (should be) covered by loan margins
- Moreover financial institution should set aside an amount equal to the [expected loss as a provision](#) ("reserves").
(Note: this is not capital buffer, this simply reduces earnings)
- Standard Credit VaR models implicitly assume independence (as in the formula above) between default, exposure, and recovery, but in reality these variables often deteriorate jointly in stress scenarios
 $\hookrightarrow$ wrong-way risk
(Example: a loan issued by a real-estate company, collateralized by commercial MBS.)

8.2.2 The key measure of credit risk: Unexpected Loss (UL) – in *default mode*

- Unexpected loss = the amount by which potential credit losses might exceed the expected loss.
 - Expected losses would have to be priced into the products (and charged to the client), unexpected losses would be covered by regulatory capital.
 - Apart from holding regulatory capital, however, many sophisticated banks also estimate the necessary economic capital to sustain these unexpected losses.
- ↪ Reserving a sufficiently high quantile of losses (denoted by VaR_α for some level $\alpha \in [0, 1]$) is viable and effective:

$$UL_\alpha = VaR_\alpha - EL$$

The Asymptotic granular one-factor model

- This model is used to quantify the minimum capital requirement related to default risk that banks must hold to operate
- Let us consider a portfolio of Loan, with Loss, see (8.5) with $T = 1$ year:

$$\mathcal{L}_N = \sum_{n=1}^N L_n \text{ with } L_n = EAD_n \times LGD_n \times \mathbf{1}_{\{\tau_n \leq T\}}.$$

- The EAD_n is the aggregation of all the exposure of obligor n in the portfolio (for loans, simply the par value). It is assumed to be deterministic.
- The default time is given as a structural threshold model, namely:

$$\tau_n \leq 1 \iff Z_n \leq -DD_n$$

The Z_n are standard gaussian random variable given by

$$Z_n = \sqrt{\rho_n}I + \sqrt{1 - \rho_n}\varepsilon_n, \quad I, \varepsilon_n \sim \mathcal{N}(0, 1).$$

$\hookrightarrow I$ is a factor representing systemic risk (economic condition).

$\rho_n \in (0, 1)$ is a parameter correlating the returns to the factor: if the factor takes negatives value the return are negatively hit.

$\hookrightarrow \varepsilon_n$ is independent of I and $(\varepsilon_j)_{j \neq n}$: it is the idiosyncratic risk (diversifiable)

- the recovery level is assumed to be deterministic and set to $\mathcal{R}$ (but conditional independence would be ok too)

- Now define: $\omega_{n,N} := \frac{\text{EAD}_n}{\sum_{i=1}^N \text{EAD}_i}$ the relative weight of obligor n within the total portfolio exposure and the overall loss per dollar of exposure:

$$\tilde{\mathcal{L}}_N := \sum_{n=1}^N \omega_{n,N} (1 - \mathcal{R}) D_n \text{ with } D_n = \mathbf{1}_{\{Z_n \leq -\text{DD}_n\}}.$$

D_n is the default indicator.

- Observe that conditionally to I the D_n are independent r.v.

- definition: The portfolio is said to be infinitely fine grained (or granular) if

$$\sum_{N=1}^{+\infty} \omega_{N,N} = +\infty \quad \text{and} \quad \sum_{N=1}^{+\infty} \omega_{N,N}^2 < +\infty.$$

$\hookrightarrow$ the portfolio is composed of a very large number of assets none of which being significant.

$\hookrightarrow$ example: if EAD_n are all the same, $\omega_{n,n} = \frac{1}{n} \dots$

- Now define

$$\tilde{\mathfrak{L}}_N = \mathbb{E}[\tilde{L}_N | I]$$

invoking an extended conditional strong law of large numbers, [13] shows

$$\lim_{N \rightarrow +\infty} (\tilde{L}_N - \tilde{\mathfrak{L}}_N) = 0 \text{ a.s.}$$

the idiosyncratic risks ε_n vanish by diversification.

$\hookrightarrow$ In practice, this means $\tilde{L}_N \approx \tilde{\mathfrak{L}}_N$ and thus

$$\text{VaR}_\alpha[\tilde{L}_N] \approx \text{VaR}_\alpha[\tilde{\mathfrak{L}}_N] =: q_\alpha(N)$$

- Now the goal is to come up with a formula for $q_\alpha(N)$. This number will be used as the value of $\text{VaR}_\alpha[\tilde{L}_N]$ in practice.
- We observe that

$$\mathbb{E}[D_n|I] = \Phi\left(-\frac{DD_n + \sqrt{\rho_n}I}{\sqrt{1 - \rho_n}}\right)$$

- We then observe

$$\tilde{\mathfrak{L}}_n = f_N(I) \text{ with } f_N(y) = \sum_{n=1}^N \omega_{n,N}(1 - \mathcal{R})\Phi\left(-\frac{DD_n + \sqrt{\rho_n}I}{\sqrt{1 - \rho_n}}\right)$$

and compute

$$q_\alpha(N) = \sum_{n=1}^N \omega_{n,N} (1 - \mathcal{R}) \Phi \left(-\frac{DD_n + \sqrt{\rho_n} \Phi^{-1}(1 - \alpha)}{\sqrt{1 - \rho_n}} \right)$$

- For the one-year time horizon, $-DD_n = \Phi^{-1}(\text{PD}_n)$ and then by symmetry of the standard gaussian law:

$$q_\alpha(N) = \sum_{n=1}^N \omega_{n,N} (1 - \mathcal{R}) \Phi \left(\frac{\Phi^{-1}(\text{PD}_n) + \sqrt{\rho_n} \Phi^{-1}(\alpha)}{\sqrt{1 - \rho_n}} \right)$$

and coming back to the dollar value

$$\text{VaR}[L_N] \approx \sum_{n=1}^N \text{EAD}_n \text{LGD}_n \Phi \left(\frac{\Phi^{-1}(\text{PD}_n) + \sqrt{\rho_n} \Phi^{-1}(\alpha)}{\sqrt{1 - \rho_n}} \right)$$

- the Unexpected loss of the first slide of credit risk, is obtained by subtracting the expected loss to the above formula taken at level $\alpha = 99.9\%$ (have in mind $T = 1$), namely

$$\sum_{n=1}^N \text{EAD}_n \text{LGD}_n \left[\Phi \left(\frac{\Phi^{-1}(\text{PD}_n) + \sqrt{\rho_n} \Phi^{-1}(0.999)}{\sqrt{1 - \rho_n}} \right) - \text{PD}_n \right] \quad (8.7)$$

- Note that in practice (regulatory capital requirement) some adjustment are made to the above formula:

$$K_{\text{IRB}} = 1.06 \sum_{n=1}^N \text{EAD}_n \text{LGD}_n \left\{ \Phi \left(\frac{\Phi^{-1}(\text{PD}_n) + \sqrt{R_n} \Phi^{-1}(0.999)}{\sqrt{1 - R_n}} \right) - \text{PD}_n \right\} \times \text{MA}_n \quad (8.8)$$

where:

- PD_n is the one-year probability of default, (internally estimated following some regulatory constraints)
- LGD_n is the loss given default,
- EAD_n is the exposure at default,
- R_n is the regulatory asset correlation = $f(PD_n, \text{asset class})$,
- MA_n is the maturity adjustment (trying to capture spread risk...),
- the factor 1.06 is a supervisory scaling factor.